

7.

IMPLEMENTING MISSING MIDDLE HOUSING

Overcoming Planning and Regulatory
Barriers

(with case studies)

IT IS CRITICAL FOR JURISDICTIONS IN CITIES LARGE AND SMALL AND CONTEXTS RURAL AND URBAN TO UNDERSTAND HOW TO ENABLE MISSING MIDDLE HOUSING TO MEET THEIR HOUSING NEEDS. THE CONVENTIONAL APPROACH to creating plans, zoning, policies, and strategies related to delivering housing needs to be rethought to effectively deliver Missing Middle Housing. Many cities have attempted to remove these barriers and fix their zoning, often with well-intended multiple area plans, zoning overlays, or design guidelines that have been less than effective. It is important to understand how to effectively introduce Missing Middle Housing into your community and then recommend changes to plans, policies, and zoning. The steps in this chapter will help you to effectively frame the Missing Middle conversation in a community, determine where Missing Middle application should be prioritized, identify barriers, and identify specific policy and zoning changes that are needed to deliver results. The chapter also offers specific zoning approaches, from simple to complex, that can be considered to overcome barriers to Missing Middle Housing.

INTRODUCE THE IDEA OF MISSING MIDDLE HOUSING INTO YOUR COMMUNITY

Creating a solid foundation for a planning process related to housing has become more and more challenging. It is important to build support early to increase the likelihood of a successful process.



Unattractive residential infill (dingbat apartment), which represents what most community members envision when the terms density, upzoning, or multifamily are heard.

TIPS:

- Identify and document existing Missing Middle types in a community to see what types of densities they output, how much parking is provided, and other basic form parameters to use as a foundation for a Missing Middle conversation.
- Think about an allowed range of Missing Middle building types, rather than density, as the starting point of zoning. Ask yourself the question: What range of Missing Middle types (predictable form) should be allowed in a particular neighborhood rather than what density (unpredictable form) should be allowed.
- Tip for council members: Ask staff to verify if a density-based planning and zoning system is required by state planning law. If it is not, ask them to provide alternative approaches.
- Address the misconception about households that rent directly. The number of households that rent is growing, even among upper-income and white households that have historically chosen to buy.¹

Framing the Housing Conversation

Framing the conversation about a community's housing needs is very important but rarely done effectively. Before launching into a discussion related to a planning or policy effort focused on delivering a broader range of housing choices, it is important to understand a community's needs and to think very carefully about how to frame or message the project to establish a solid, attack-resistant, support-building armature.

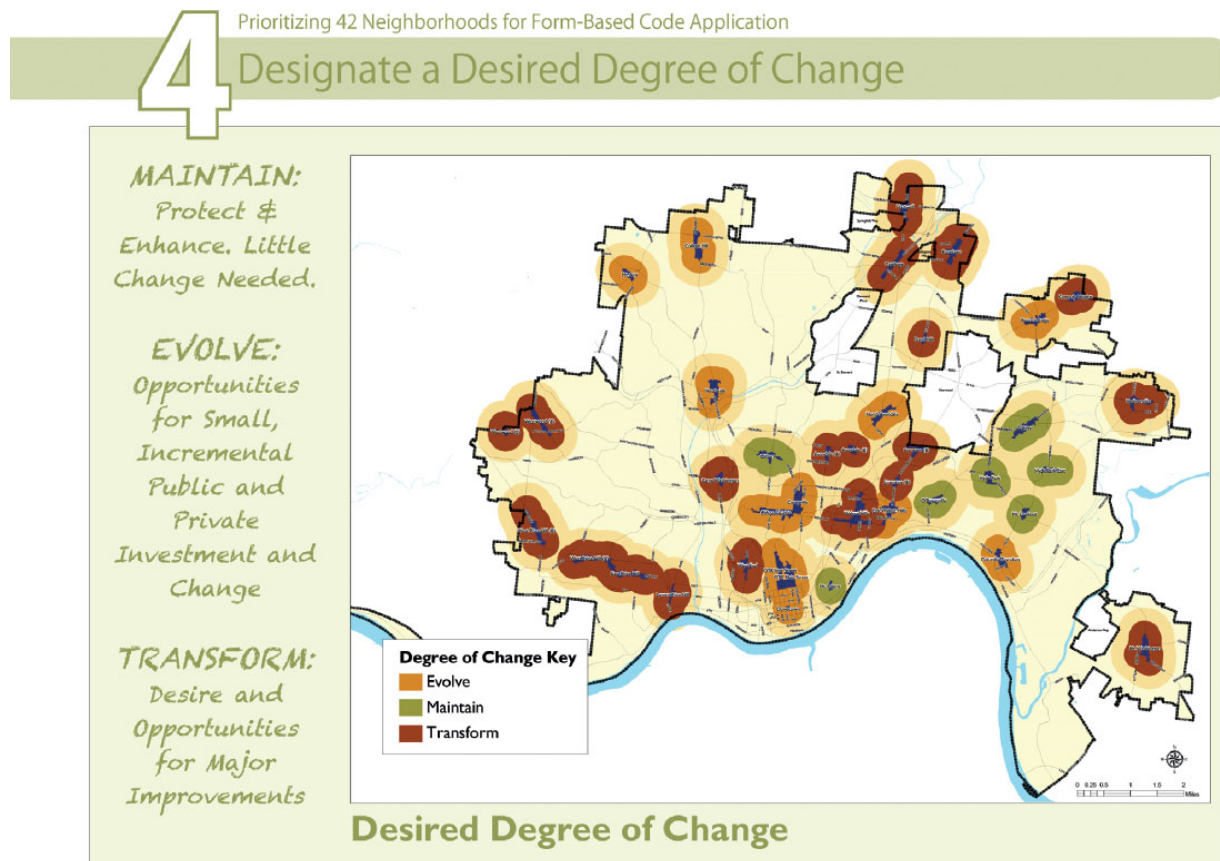
The housing conversation is often framed around needing more density or more multifamily housing, or even needing to upzone neighborhoods to deliver more housing. It is very difficult to build support for an effort framed in this way because of widespread negative perceptions associated with these terms, often due to undesirable results from past efforts. Therefore, the terms *density*, *multifamily*, and *upzoning* should be avoided altogether when framing the conversation or intent of a planning process.

Using the term *Missing Middle Housing* and the names of the types, such as *duplex*, *town house*, *courtyard apartment*, *cottage court*, and so on is an effective way to get people to think about a form and scale of buildings. Convey that Missing Middle is primarily about form and scale (house scale) and secondarily about middle income. Not vice versa. Reinforce that it doesn't matter whether these types are for sale or for rent. A single-family house, whether it is rented or owned, is still a single-family house.

Instead of using the terms *density*, *multifamily*, and *upzoning*, use these terms in framing the housing conversation:

- Missing Middle Housing (focus on form, scale, and type)
- Duplex, fourplex, cottage court, courtyard building, mansion apartment
- House-form or house-scale buildings that just happen to have multiple units within them
- Housing choice
- Attainability (where will the good teachers, firefighters, police, and so on live in your community? Where will your children live?)
- House-scale buildings with multiple units within them

- Broad range of housing types



This map from Cincinnati's award-winning comprehensive plan labels each identified walkable urban place type with a desired degree of change, thus establishing a form-based policy.

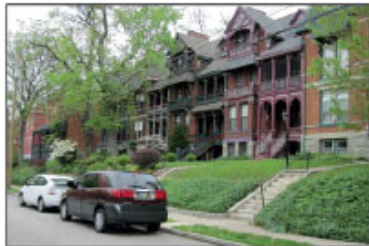
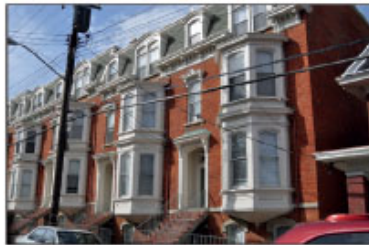
Clarify the Desired Degree of Change

When framing a housing conversation in your community, be clear about the desired degree of change as identified in the comprehensive plan or in the small-area plan or for each sub area or neighborhood of a city. The desired degree of change in each area should be based on thorough analysis, input from the community, clearly defined goals and objectives, and, ideally, a detailed vision. The reason this is important is that most community members typically envision the transformative approach with widespread and dramatic change everywhere, but the reality is that many planning efforts are really targeted for more incremental change.

Townhouse: Detached



Townhouse: Attached



Mansion Apartment/Apartment House





Building Types

Cincinnati, Ohio
April 28, 2012



Sample of a poster that gives a good visual overview of Missing Middle types by category and is an invaluable resource in building support for Missing Middle Housing.

Use the Following Degrees of Change

1. *Maintain*: Smaller, more incremental changes, mostly reinforcing the existing scale of an area
2. *Evolve*: Opportunities for small-to-medium-sized public and private investments or projects. Minor changes in scale. Opportunity sites should be targeted.
3. *Transform*: Opportunities for larger-scale changes, such as a significant increase in scale and possibly mix of uses. The changes are more likely to be widespread and not on focused sites.

The reason this is important in framing is that most community members typically envision the transformative approach with widespread and dramatic change everywhere, but the reality is that many planning efforts are really targeted for the evolve degree of change, that sensitively allow for smaller changes, especially if Missing Middle Housing is the focus or priority.

Gather Local Examples of Missing Middle Housing

In this preliminary analysis phase, collect photographs of local examples of Missing Middle types. If local examples are not available, use regional examples. Typically local planners, community members, or historians know where these examples exist. If not, local examples can be found using Google Maps (see [box 7-1](#)), GIS data, or data from your county assessor's office to create a preliminary set of maps that illustrate the number of units per lot for targeted neighborhoods. These photos ideally would be arranged as posters or in an online database organized by housing type and referenced and utilized frequently throughout the process. This step seems rudimentary, but it should not be skipped. Photos reinforce the shift away from density to a conversation about housing type, scale, form, and who is

going to live in these homes. In addition, some community members will refer back to these photos throughout the process, even as more detailed information is produced, because they can relate to these local photo examples more easily. The website MissingMiddleHousing.com presents a large palette of photographic examples of each Missing Middle type for this purpose.



Walking tours are often part of the programming for a series of events to effectively introduce Missing Middle Housing to a community.

Host an Introductory Missing Middle Housing Presentation or Event

A presentation is an important tool for explaining Missing Middle Housing. In this early phase, presentations are typically focused on the general public or local or regional planning groups and are supplemented with meetings with decision makers such as the city manager, mayor, council members, or

planning commissioners. Focused discussions and short presentations are also hosted for groups such as the local development community and affordable-housing advocates.

A guided Missing Middle walking tour will ideally supplement the introduction of Missing Middle Housing to a community. This allows people to see how existing examples function and to build a better understanding of their form and scale and is an effective step in building support from community members. The tour should be led by a Missing Middle expert, with additional guides who know the local zoning, as well as longtime community members who know the history of the neighborhood.

The intent is to build an initial understanding and value of Missing Middle Housing and lay the foundation for a positive discussion and begin to build support from local leadership.

Sometimes this effort is funded by a jurisdiction; other times it may be partially or completely funded by local stakeholder groups that are interested in moving the housing conversation along. For example, in Bend, Oregon, the process was initiated by Building a Better Bend. In Arlington County, Maryland, an initial presentation was sponsored by Alliance for Housing Solutions. In Greenville County and the City of Greenville, South Carolina, a group of realtors and members of the development industry called Impact Greenville found funding for the entire process. In other instances organizations like AARP have initiated the Missing Middle conversation in communities, hosting a road tour through Southern Oregon.



This cottage court was designed for a city-owned infill site as part of the award-winning Richmond Livable Corridors project in Richmond, California, as part of a strategy to deliver more affordable, high-quality housing choices to existing residents.

DETERMINING WHERE TO APPLY MISSING MIDDLE HOUSING

Regardless of the size of a city or county, there is always a hierarchy of walkable urban places that can be considered for Missing Middle application. The following are context types to be considered.

Small, Individual Lot Infill

This approach tends to make the most sense in pre-1940s neighborhoods that are already walkable, mostly built out, and often have a mix of Missing Middle types. Start by identifying where these types exist and the barriers to building these types in your current parking and engineering standards, and then revise or refine your zoning and other regulations to enable these types. Often existing Missing Middle types would not be allowed to be constructed again according to current regulations. This could be part of a small-area planning effort as well.

Secondary Corridors

Many cities have zoned miles of corridors for commercial uses, and many of these corridors are trapped in a state of disinvestment as the commercial uses moved farther out of the city or back into the downtown cores. These corridors, which are often abutted by single-family or mixed-type neighborhoods, are great opportunities to enable Missing Middle Housing. These are typically secondary corridors or corridors with shallower lots that could not easily accommodate larger mixed-use or higher-density housing but are ideal for lower-intensity Missing Middle types.

Transitions from Higher-Intensity Corridors

Many cities over the past couple decades have introduced strategies, policies, and zoning to allow higher-intensity development, often transit oriented, along their major corridors. The result has often been awkward, with five-plus-story buildings abutting single-family homes, which usually results in an outcry from adjacent neighborhoods. Applying Missing Middle Housing is a great strategy to transition from these corridors into lower-scale neighborhoods.



Secondary corridors with vacant and underutilized properties are prime opportunities for Missing Middle-scale infill projects.



The scale of Missing Middle types makes them perfect for providing transitions from higher-intensity corridors into single-family or lower-intensity neighborhoods.



Proposal for North Berkeley BART with four-to-seven-story buildings in the center and Missing Middle types at the edges to transition in scale into adjacent single-family neighborhoods while still achieving an average density over the targeted 75 du/acre.

Thoughtful Transit-Oriented Development (TOD)

The application of Missing Middle Housing around new or existing transit stations is a good strategy where the lots are smaller or constrained, making

larger buildings difficult, or where single-family homes abut the stations. Missing Middle Housing should be considered in some transit-rich locations as an alternative or as a transition from the larger buildings directly adjacent to stations into neighborhoods. Not all transit-oriented development needs to be five-plus stories.

Interface of Greyfield Sites with Adjacent Neighborhoods

Rapid change in retail has left many mall sites, including medium to large strip malls, vacant in cities across the country. These are prime opportunities to develop new walkable communities, which may have larger, mixed-use buildings as a component along the corridors, but an effective strategy can also integrate Missing Middle Housing within blocks at the edges of the site to transition into adjacent neighborhoods.



Blocks composed of Missing Middle types provide great transitions in scale at the edge of larger, redeveloping greyfield sites that also have higher-intensity buildings along the corridors.

Downtown Transitions (Small-to-Mid-Sized Cities)

The edges of most downtowns suffer from poor planning and ineffective zoning. Many of these areas had a great mix of Missing Middle Housing types that were torn down for freeways, parking, or suburban-style development such as drive-through banks.

New Growth Areas at the Edge of Cities

Many cities and counties have targeted undeveloped areas to accommodate growth and new development. If they enable or require compact, walkable urban development as part of their long-term planning, Missing Middle Housing will be a critical component. As construction costs have risen, builders are finding that Missing Middle Housing types can deliver housing at better price points than single-family housing, especially for entry-level buyers.

Single-Family Neighborhoods

This is the most challenging application politically, but this is an area where we are seeing change, such as the sweeping legislation in the City of Minneapolis to ban single-family zoning. The changes are in response to the extreme level of the housing crisis in some areas and the fact that a majority of land has been zoned for single family in many cities. Cities are thoughtfully considering allowing Missing Middle types to deliver more housing choices and achieve a higher level of affordability.

IDENTIFY BARRIERS

The following steps will help to develop a solid foundation for future planning efforts, specifically a comprehensive plan or general plan update and detailed, targeted zoning fixes to enable Missing Middle Housing.



Many buildings on lots adjacent to downtowns in most small- and medium-sized towns were torn down and are now parking lots or one-story suburban buildings that make these sites prime opportunities for Missing Middle infill within a one-block walk to downtown amenities.

Review the Comprehensive Plans (City, County, or Both)

Review the comprehensive plan to identify policies that present barriers or limits to achieving Missing Middle Housing. This is most often done at a citywide scale, but in regions where there is a lot of growth in the surrounding county or counties, it often makes sense to analyze both the city and the county comprehensive plans. This process should assess where the multiunit land-use classification is designated, if it is proposed in logical locations, and if the defined geographic area is large enough to accommodate non-single-family housing choices if the barriers were removed. The review should focus on the future land-use categories and their location and parameters, such as allowed densities and lot sizes. Then policies should be identified and assessed related to delivering more housing choices. This analysis can help jurisdictions to make targeted

changes in future land uses and where they are mapped. It can also guide the refinement or addition of policies to remove these barriers rather than taking on the time-consuming work of a complete comprehensive-plan update.



Comparison illustrating the gap between the low densities allowed by existing future-land-use categories and higher densities required to enable Missing Middle Housing by type in Greenville County, South Carolina.

Barriers for MMH in Greenville County											
Barriers to MMH	Imagine Greenville Comp Plan	Berea Plan	Brandon Plan	Conestee Plan	Monaghan Mill Plan*	Cherrydale Plan	City View Plan	Dublin Road Plan	Dunean Plan	East Woodruff Plan	Fountain Inn Plan
Max. Density Allowed: Too Low	●	●	?	?	●	●	●	●	?	●	●
Min. Lot Area: Too High	●	●	●	●	●	●	●	●	●	●	●
MMH Types Identified ¹	●	2	2	2	2	2	2	2	2	2	2
Content is Supportive of MMH	●	●	●	●	●	●	●	●	●	●	●
Notes: ¹ Only reference to duplex, townhouses and multifamily. ² Plan update proposes down-zoning from multifamily to single-family, and down-zoning R-7.5 to R-10. ³ Policy direction to limit “apartment” style of multifamily housing leaves it unclear if MMH is intended or not. ⁴ Majority of area is zoned for single-family. ⁵ The future land use map includes a TND category that allows mixed residential types within a maximum overall density of 1.7 units per acre for the master planned area. ⁶ In R-M and R-MA, multifamily requires a min. of 2-acre site size and 12,000 sq. ft. size respectively. ⁷ In R-M, once 2-acre site is provided, no individual lot minimum. ⁸ Duplexes through Fourplexes mentioned but majority of R-M Zoning being changed to Single-Family Zoning. [*] In Progress version reviewed The PD and FRD are not shown because it is unclear what they allow given that each project is reviewed on a case-by-case basis.											
Key: ● Barrier ? Unclear ● Not a Barrier											

This table summarizes the analysis of the twelve small-area plans that the City of Greenville has completed in recent years, including the clarity of intent related to delivering Missing Middle Housing and if they remove zoning barriers.

Review the Small-Area Plans

Cities often have plans for smaller geographic areas within their jurisdictions, such as neighborhoods, corridors, transit areas, or downtowns. Unfortunately, they are not always clear on the intent, often not getting beyond general policies, such as “allowing a broader range of housing choices,” without clarifying the scale or type of housing allowed for that specific area. In addition, these plans often do not identify zoning changes that need to be made to deliver the plan’s intent, and sometimes introduce design guidelines or design parameters that are conflicting or inconsistent with the existing zoning. This step assesses these plans and identifies the specific gaps and barriers within them that prevent the effective delivery of Missing Middle in the planning areas.

For example, Greenville, South Carolina, has completed twelve small-area plans over the course of fifteen to twenty years, and only a few of the most recent efforts were clear in their ultimate intent beyond “deliver more housing choices.” Of these twelve plans, only two of the more recently completed plans, which were implemented with form-based codes, had clear intent and had truly eliminated the zoning barriers. Unfortunately, this type of disconnect between intent and application is fairly common. The form-based codes that were created were all one-offs, created specifically to be applied to a limited area rather than being part of a comprehensive strategy for zoning changes that is typically recommended. But they do provide a model for future coordinated changes.

Review Zoning

A targeted group of residential zoning districts, focusing on medium-density or multifamily zones, should be analyzed to assess if they can effectively deliver Missing Middle Housing. In the preliminary assessment, identify the top barriers and communicate them in a clear, graphic form.

Zones that truly enable Missing Middle are often missing entirely from a zoning code. Does the city have a zone that allows a minimum of 35 and ideally up to 75 du/acre residential uses but also caps the height at two to two and a half stories? If so, does it regulate a maximum width of building to 45–60 ft.? The zoning codes for 99 percent of cities across the country do not have a single zone that meets all of these requirements and thus truly enables Missing Middle Housing. Typically the biggest barriers are allowed building envelopes much larger than Missing Middle (height, width, depth), densities that are too low, minimum lot sizes that are too high, and parking requirements that are too high. Often there are single-family zones that enable duplexes, often as single family attached, and then the medium-density/multifamily zones immediately jump up to allowing a 45–50 ft. tall building with unlimited width as long as it meets the density limitations.

TIP

A good supplemental test is to pick a neighborhood, map the number of units per lot, then verify conformity or whether the existing development on these lots meets the existing minimum lot-size requirements, maximum density, or maximum square footage required per lot using a system similar to the micro-scale analysis process.

MAP PRIORITY AREAS FOR MISSING MIDDLE HOUSING APPLICATION

The areas for Missing Middle Housing may have been identified generally at the very beginning of the process. Now is the time to identify and map the specific areas where Missing Middle Housing application should be prioritized. It is recommended that this step be completed before starting a comprehensive plan update to better inform that process.

Examples include the areas within or surrounding a downtown core, downtown transition areas (a prime opportunity for Missing Middle), neighborhood main streets, areas near parks and open space, and other areas that deliver walkability. These areas are typically identified through careful analysis of proximity to schools, parks, commercial services, transit, and other amenities and should be represented on a map using the five-minute-walk or pedestrian shed. It is very important to focus the discussion in the short and long term about where it makes the most sense to enable Missing Middle Housing within a jurisdiction and to establish this foundation for future planning efforts.

The map that is generated from the analysis can function as the foundation for comprehensive-planning updates, in particular for a future, place-based land-use map, transit or other mobility-network improvements, and prioritization of other city investments in these areas.

The areas that are suitable for Missing Middle Housing may represent a small or a large percentage of the jurisdiction overall. If it is a large percentage of the area, the challenge can be prioritizing the opportunities to enable Missing Middle. If it is a limited geography defined that is appropriate for Missing Middle, then the challenge is often how to transform additional targeted areas to improve walkability and enable Missing Middle Housing to have enough impact. The latter is more likely the case in a jurisdiction that developed mostly or completely after the 1940s.

Box 7-1

Where and How to Find Missing Middle in Your Community Using Google Maps

The following steps can be followed to use Google Maps to help identify existing Missing Middle Housing in a community. This can supplement the mapping and the walking tour.

Step 1: Open Google Maps for the city you are analyzing and turn the Map View option on (Satellite View off).

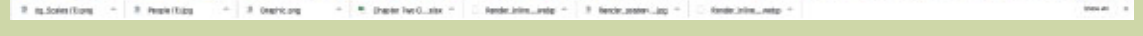
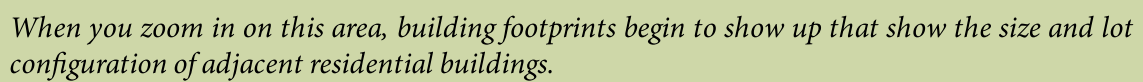
Step 2: Find neighborhood commercial areas/main streets: Start in areas around pre-1940 commercial districts, which were often neighborhood main streets. With Map View on, these areas will show up highlighted in yellow. You may want to use a zoning map to see where smaller geographic areas are zoned commercial to understand where to start or test.

Step 3: Search adjacent neighborhoods Missing Middle types: Turn on the Satellite View and then zoom in on one of the areas that is adjacent to the neighborhood commercial district. Look for multiple rooftops, courtyard configurations, or building outlines that are slightly bigger than others that likely will be Missing Middle types. This is the challenging part that will take some time and training.

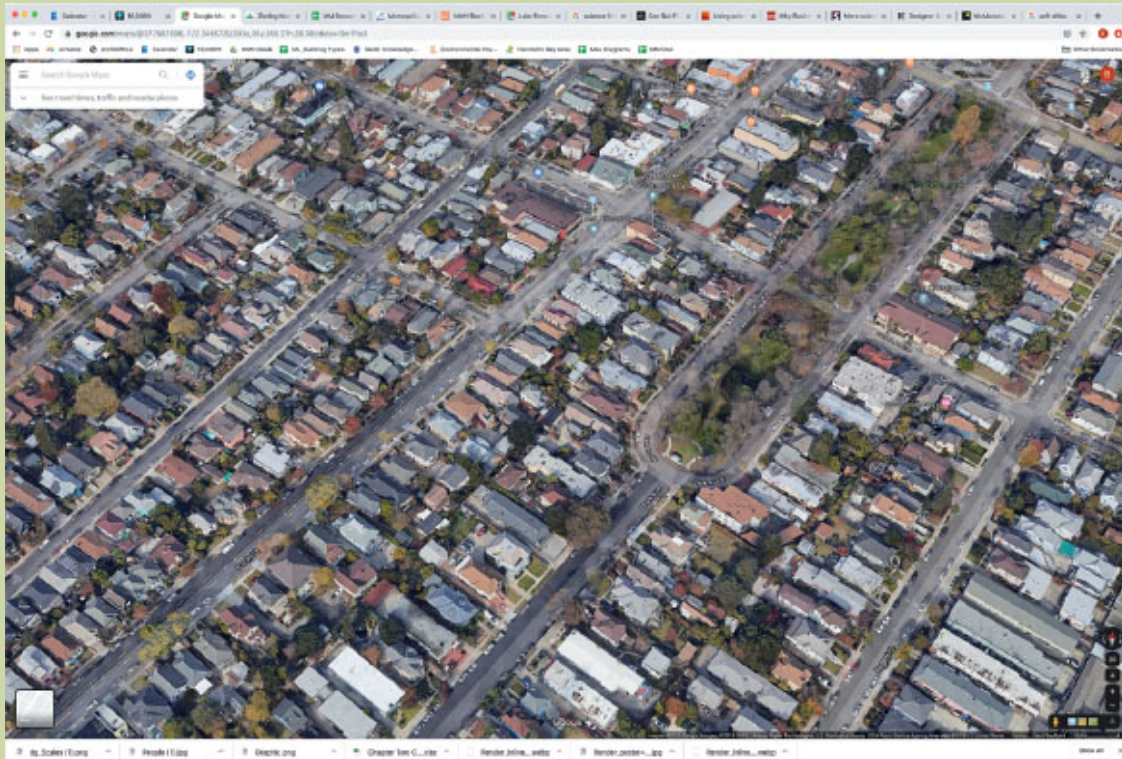
Step 4: Zoom in and out and use 3-D View and Street View to verify potential types: When you have found a building you think is a Missing Middle type, click on the 3-D View button next to the + -Zoom tool to confirm whether or not the lot and type you have chosen is a Missing Middle type. Sometimes you may need to drop into Street View and look for multiple doors, mailboxes, doorbells, to confirm this.

Step 5: Take a screenshot of the street view for your records and be sure to include the address at the top left corner. These screen shots will be your catalog of potential Missing Middle types to visit. They can also be compiled into one document to serve as an initial overview of existing types in a community that will ultimately be replaced with better photos.

Step 6: Create a map in My Maps and pinpoint each of the types you find for future reference.



When you turn the satellite option of the map on, you can begin to look for slightly larger buildings or courtyard configurations. This takes some time to train yourself to find these variations and Missing Middle Housing.



This search for Missing Middle buildings is often easier in 3-D view, which can be turned on by clicking the 3-D button in the lower right of the view.



When you find a building that has a footprint that looks slightly larger than a single-family home or appears to be a courtyard, you can go into street view to verify by placing the yellow person in the lower right onto the map to see if the building has multiple units. This unit has two doors, thus having at least two units. You can go back and forth between the aerial, 3-D view, and street view to find Missing Middle types.

Where are Greenville's walkable centers?

The map identifies walkable environments in the City of Greenville focused around a variety of walkable centers identified through this analysis. The walkable environments shown represent approximately 2.15% of Greenville.



Key

Identified Walkable Centers

- Downtown Core
- Urban Center/ Downtown Transition
- Neighborhood Main Street
- Medical/Institutional

Amenities

- Parks/Open Space

Walkable Environments

- 5 min walking distance from Centers
- 5 min walking distance from Bus Stops

Zoning Districts

- Residential Districts
- /// Commercial Districts

This map defines walkable contexts that are priorities for Missing Middle Housing application. This map can be used to inform the comprehensive and small-area planning and to prioritize city investment in mobility improvements and other city expenditures for these areas.

RECOMMEND CHANGES TO PLANS, POLICIES, AND ZONING

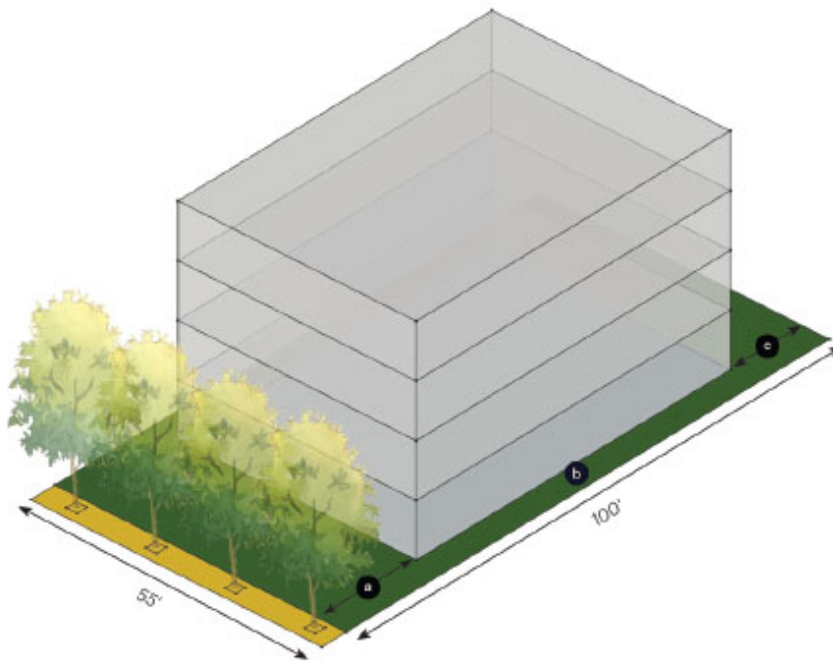
The intent of this step is to utilize the previously completed analysis of policies, planning, and zoning to complete another level of deeper analysis to inform specific recommended changes to policy, plans, and zoning.

Graphically Test Existing Zoning to Inform Specific Changes

Using the map created in the previous step, select a series of small-to-medium-sized lots within the areas identified as priorities for Missing Middle Housing. The lots selected should represent a range of typical conditions such as lot size, adjacencies, alley-loaded or not, and so on. They are used to test the existing development standards (zoning) and design guidelines on these lots. The goal is to create a detailed, hypothetical, three-dimensional build-out on a variety of lot sizes that clearly illustrates the issues identified in the regulations on a variety of different lot sizes and conditions. For example, such a 3-D graphic often shows how limiting parking requirements are for delivering multiple units, that large side setbacks discourage any development, and that compatible-scaled buildings can deliver much higher densities than the zoning regulations allow on small lots. This hypothetical development scenario analysis informs the recommended changes and refinements in the zoning, area plans, and comprehensive plan policies to effectively enable Missing Middle Housing.

R-M2 Zone - Development Standards

Max. Envelope per Existing Standards



Left: The diagram illustrates the R-M2 zone's maximum development envelope.

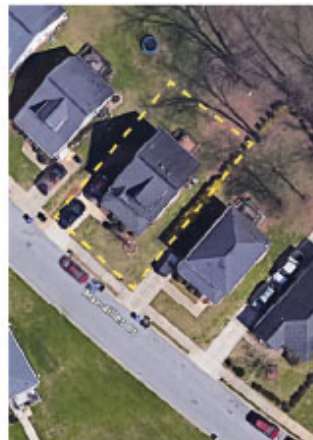
Key

Setbacks

- a** Front = 20'
- b** Side = 5' (min. 15' for multifamily)
- c** Rear = 15'

Below: Example of 55' x 100' lot and existing development.

R-M2 Zone, 55' x 100' Lot Size Max. Envelope per Existing Standards	
Building Form	
Lot Area	5,500 sf
Max. Height	40'
Max. Lot Coverage	40%
Parking	
Min. Parking Spaces	2 per unit for single family; 1.5 per unit for multifamily
Density	
Max. Allowed Density	20 du/ acre



Existing zoning standards hypothetically allow up to a maximum of 40 ft. of height and 20 du/acre.

R-M2 Zone - Allowed

Max. Potential Development per Existing Standards

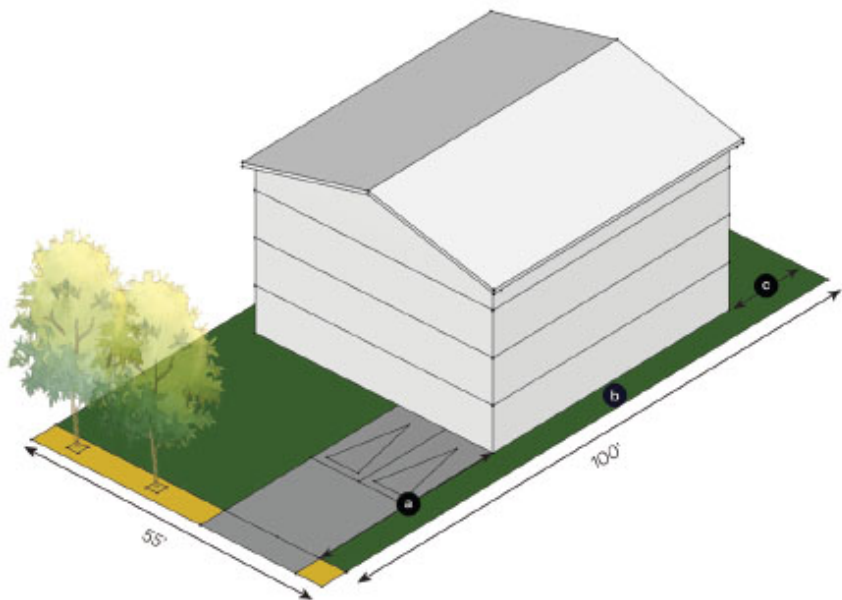
Right: The diagram illustrates what type of development and form are possible after applying all standards.

***Note:** After applying the minimum side setbacks for multifamily buildings, the resulting building is 25' wide, significantly limiting the development of the site to essentially a townhouse.

Key

Setbacks

- a** Front = 36'
- b** Side = 5' (min. 15' for multifamily)
- c** Rear = 15'



Note: Although the building complies with the zone standards, the resulting building is out of scale with the neighboring buildings and does not contribute to a streetscape of building facades.

R-M2 Zone, 55' x 100' Lot Size Allowable Development per Existing Standards	
Building Form	
Resultant Footprint	2,200 sf
Resultant Height/ Stories	35/ 3
Resultant Lot Coverage	40%
Parking	
Min. Spaces Required	2
Density	
Number of Units	1*
Resultant Density	8 du/ acre



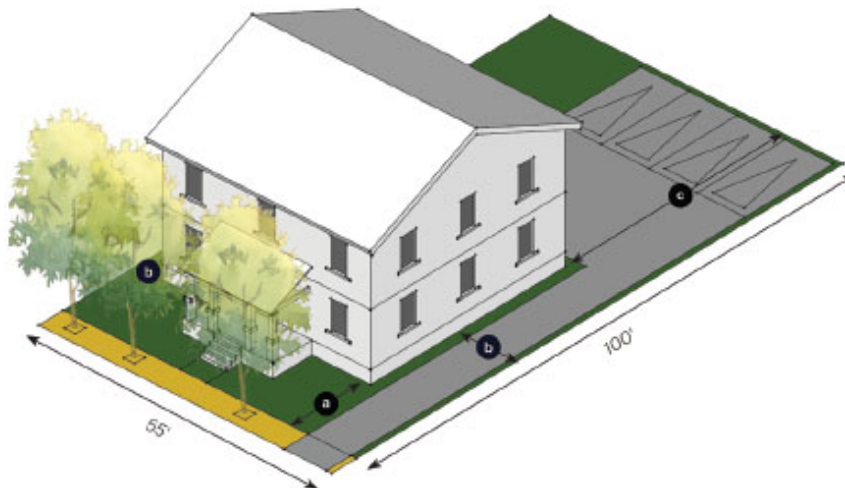
Even though it is in a multiunit zoning district, the on-site parking requirement of 1.5 spaces per multifamily unit and the low density cap of 20 du/acre encourage the site to be built out with one large single-family home that generates a density of 8 du/acre, thus not achieving the intent of the multiunit zone.

Graphic analysis of R-M2 zone on a 55 x 110 ft. deep infill lot.

R-M2 Zone - Missing Middle

Option 01 - Fourplex

Left: The diagram illustrates how a fourplex can fit on the lot while fitting in the pattern of buildings in the area.



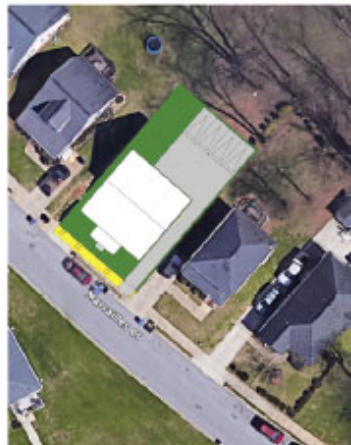
Key

Setbacks

- a** Front = 15'
- b** Side = 5' (left), 11' (right)
- c** Rear = 45'

Below: The fourplex provides a small front yard and a building footprint smaller than what the R-M2 zone allows.

R-M2 Zone, 55' x 100' Lot Size Missing Middle Option 01	
Building Form	
Resultant Footprint	1,560 sf
Resultant Height/ Stories	32' / 2
Resultant Lot Coverage	28%
Parking	
Spaces Required	4
Density	
Number of Units	4
Resultant Density	32 du/ acre



A hypothetical build-out of the same lot of a two-story fourplex, which maxes out the realistic Missing Middle-scale development potential on the site using a requirement of one parking space per unit on-site. It generates a density of 32 du/acre. This build-out scenario is used to recommend adjustments in the current zoning to encourage Missing Middle Housing infill including decreasing parking requirements to one space per unit and increasing density to 32 du/acre. Note this build-out has one and a half times the density of what the current zone district allows, but it only needs to

utilize two of the four stories allowed, and 28 percent of the allowed 40 percent lot coverage, to achieve this density. This is an indicator that the development standards are ineffective.

Inform Comprehensive Plan and Small-Area Plan Land Use and Policy

The addition of this detailed lot testing and the analysis done previously can inform specific changes to comprehensive plans and small-area plans to effectively deliver Missing Middle Housing. The following is a list of typical policy changes:

- Define walkable environments and create a map that identifies them as priorities for Missing Middle Housing application and differentiates them for other policy applications as well, such as reduced parking.
- Define a specific intended physical form and character for these areas.
- Define Missing Middle Housing as house-scale buildings with multiple units within these plans.
- Clarify the intent of Missing Middle Housing to be both for small-lot infill and within new communities.
- Introduce the full range of Missing Middle types and their characteristics, ideally using photographs of local examples.
- Be clear that Missing Middle Housing is different than conventional multifamily development.
- When promoting housing choices, be clear on the scale, form, and type of housing that is desired in certain areas.
- Define a desired degree of change (maintain, evolve, transform) for each neighborhood/sub area and clearly articulate how that informs Missing Middle application as well as future land use and zoning. For example, is the intent to completely transform a single-family neighborhood into a neighborhood with a mix of Missing Middle Housing or to allow a neighborhood already zoned for multifamily to evolve in a more incremental manner?

- Create land-use categories that specifically identify the desired intent of Missing Middle Housing that allows higher densities but requires smaller buildings.
- Option 1: Future Land Uses
 - High density: house scale
 - Medium density: house scale
- Option 2: Future Land Uses
 - Missing Middle Housing
 - Upper Missing Middle Housing

ZONING FIXES

As one of the major barriers to the delivery of Missing Middle Housing, zoning needs to be thoughtfully fixed and in some instances replaced. Several strategies, both targeted and comprehensive, are summarized in this section.

Address Nonconformities of Existing Missing Middle Buildings

Many good examples of existing Missing Middle Housing types are nonconforming, which means the existing zoning regulations—often setbacks, lot size, density, or parking requirements—would not allow you to build them today. This often makes it hard for an owner to maintain the building without having to comply with the newer regulations. This compliance with regulations is typically financially infeasible so the buildings are often left in a state of disrepair because the owner has been put in an impossible position.

First, identify these nonconformities in existing Missing Middle types as legal nonconforming buildings, lots, and uses. Ensure that there is a section in your code that enables the owners of these nonconforming buildings to make improvements, to any dollar amount, as long as they do not add square footage to a building, without having to comply with current size, use, density, parking, or form standards. This will enable (but not require) the owner to more easily maintain the Missing Middle buildings so they are a welcome part of a community and not the eyesore.

Targeted Missing Middle Housing Zoning Fixes

The following are quick, targeted changes and refinements that can be made to your zoning that will not typically get a lot of pushback so you can immediately remove barriers for the delivery of Missing Middle Housing.

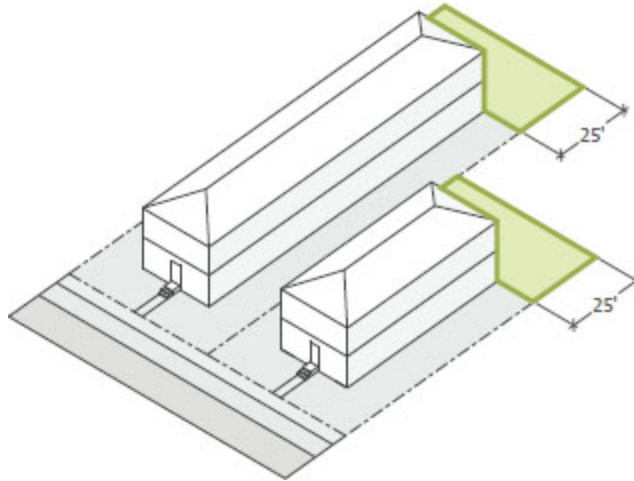
1. Regulate maximum building width and depth (keep the scale small).
2. Increase maximum allowed density or reduce minimum required square footage of lot per unit.
3. Adjust maximum allowed heights to be no more than two and a half stories.
4. Reduce minimum-lot-size requirements and replace with minimum lot width.
5. Reduce or remove parking requirements.
6. Remove open-space requirements.
7. Map the zones that allow these types more broadly.
8. Do not allow multiple single-family homes on a lot or tuck-under town houses.

Regulate Maximum Width of Building to Be Approximately 45–60 Ft.

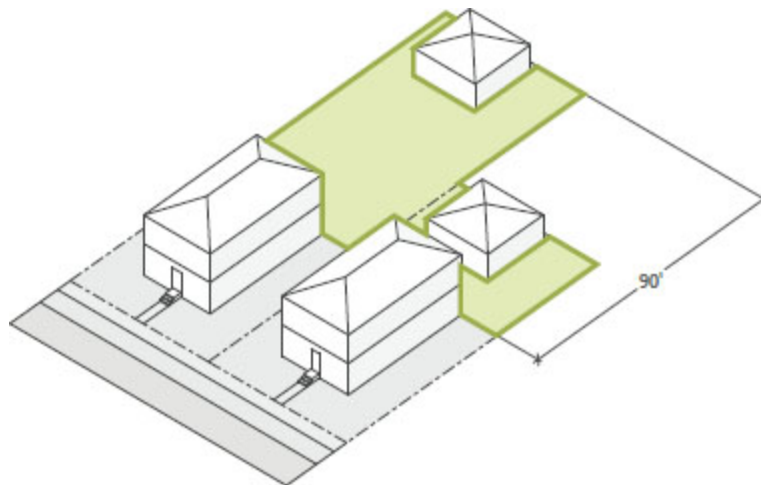
Why is this important? This ensures that if a lot size gets bigger, or if someone aggregates a couple of lots, the building does not get larger than what is desired, but rather two, appropriately scaled buildings get built. When lot sizes vary, or lot aggregation occurs, it is often when a density-based system falls apart and delivers out-of-scale buildings. This regulation of maximum building width assures community members that buildings that get built, even if density is increased, will be house scale.

TIP:

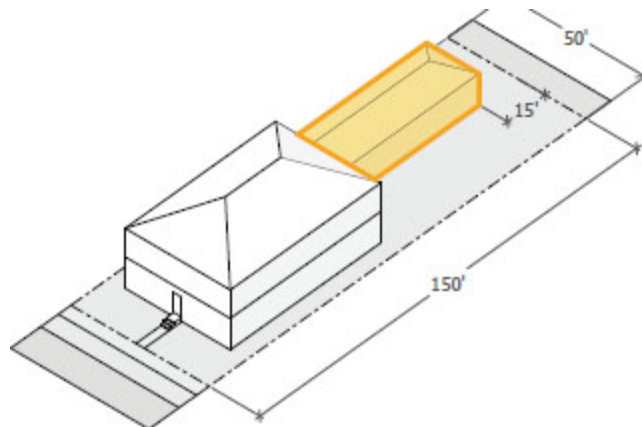
You may have one zone that regulates a maximum building width of 45 ft. so that the form is more compatible with single-family homes, and then another zone that allows the same range of types but regulates a maximum building width of 65 ft. and is more thoughtfully mapped along evolving corridors or as transitions from corridors into neighborhoods. This is a nuanced, form-based approach to allowing Missing Middle types.



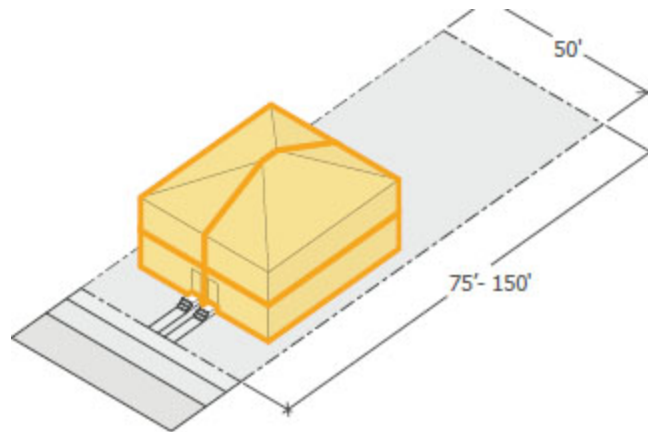
Rear setbacks are ineffective in defining a compatible built form. If lot depths change on adjacent lots, a much larger, incompatible form is allowed.



Maximum building-depth regulations effectively regulate a house-scale form, even when lot depths change. This is also why floor-to-area ratio (FAR) regulations are ineffective with varying lot sizes.



Some communities are okay with a building going deeper into a lot, as long as the element or wing of the house that goes deeper is only one story.



This is a fourplex building form that represents the maximum scale that should be allowed in a Missing Middle zoning district. The exact same fourplex building form generates a broad range of densities, FAR, and lot coverage percentages when applied to different lot sizes, thus illustrating the need for a form-first approach to zoning rather than using these numeric parameters to enable predictable results and to remove barriers.

Regulate Maximum Depth of Approximately 45–60 Ft. Deep

Regulating a maximum depth of building is critical for Missing Middle Housing types. Unfortunately, depth is rarely regulated. It is a more direct and effective way to regulate a predictable form than relying on rear setbacks. The problem with a rear setback is if a lot gets deeper than the one adjacent to it (which often happens in pre-1940s neighborhoods that have irregular lot patterns) then the scale and form of a building is allowed that is not compatible with the one adjacent to it. A one-story wing is sometimes allowed to project farther into the rear yard if it is thoughtfully regulated, as per the figure above.

Ensure House-Scale Buildings with Medium to High Densities

The most basic strategy related to effective zoning for Missing Middle Housing is to ensure that zoning has development standards (within one or more zones, as defined below) allowing medium to high densities but requiring house-scale buildings/forms. It is this regulated form/building envelope—not density, FAR (floor-area ratio), or other numeric parameters

—that delivers predictable built results. This has been done in jurisdictions across the country, including Flagstaff, Arizona; Cincinnati, Ohio; Iowa City, Iowa; and Kauai County, Hawaii.

Increase Maximum Allowed Density or Reduce Minimum Required Square Footage of Lot Per Unit (If You Must Use Density Requirements)

It is unnecessary to regulate density if you carefully regulate the building form/allowed envelope. But if a density requirement is politically necessary, make sure the maximum allowed density is high enough: at a minimum, 35–45 du/acre and as high as 75 du/acre to truly encourage smaller units. Use the tables in [chapter 5](#) provided for each of the types to understand the range of densities each generates on typical lot sizes.

TIP:

If you need to keep density requirements, start thinking about density as output (what a specific type on a specific-size lot generates) versus the input or primary controlling factor. You can compare the typical densities that the Missing Middle types generate to the densities allowed by your current zoning districts.

Adjust Maximum Allowed Heights to Allow No More Than Two and a Half Stories

This is the component/regulation that is often missed but is critical to delivering house-scale buildings. The difference between two and three stories in terms of perception is big. The upper end of Missing Middle is often three to four stories, and cities should have another zone that enables these taller, Upper Missing Middle buildings, but a zone that regulates the core of Missing Middle should not allow buildings taller than two to two and a half stories. It is important to note that this three to four stories only pertains to certain types, not all of the Missing Middle types. If more height is allowed within a zone, it also often overinflates land prices, thus making it impossible to build, even if a builder wants to deliver Missing Middle.

Box 7-2

Create a Walkable Neighborhood or Missing Middle Housing Overlay

If you can't add new zones and need to tweak existing zones, you can apply the new standards and densities to the existing zone through a targeted walkable neighborhood overlay district. To implement the walkable neighborhood overlay, define areas on the zoning map where you want the new regulations to override the existing ones, then define the new regulations, such as density, heights, parking, building width and depth, that replace the existing regulations. This enables the new standards to be used in certain walkable areas while not changing zoning elsewhere. Overlays should typically be avoided because multiple overlays are often hard to understand and administer, and they are often used as the easy fix, when a more comprehensive one is what is actually needed for long-term effectiveness. That being said, a thoughtfully created overlay can be a good code hack to enable Missing Middle. You need to be sure the new standards do not conflict with existing standards by clearly stating that these new regulations trump preexisting ones. Make sure the overlay exists on your zoning map and can be found easily within your zoning document.



This photo shows the dramatic difference in perceived scale between a two-story and a three-story building. This is why most Missing Middle types max out at two to two and a half stories.

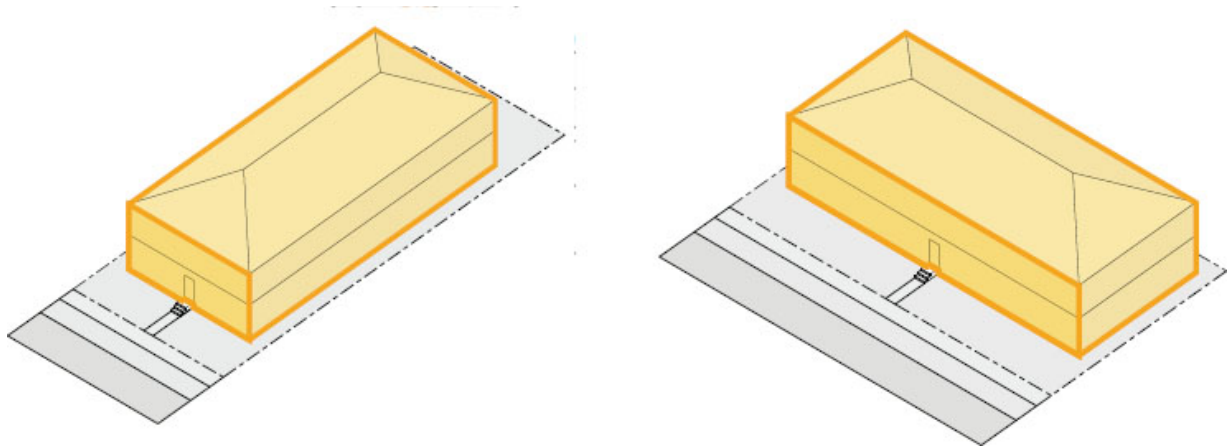
TIP:

If a zone currently allows more than two to two and a half stories, you need to be careful not to downzone or reduce development potential without careful consideration or policy direction. This is not done very often, and we do not typically recommend it. You will need to find areas that are not already zoned for larger-scale buildings, often meaning selecting targeted areas currently zoned for single family or converting areas zoned for commercial to all Missing Middle.

Reduce Minimum-Lot-Size Requirements and Replace with Minimum Lot Width

Make sure Missing Middle types are enabled on one existing, typical lot width. Zoning codes typically regulate a minimum lot size. A more effective way to regulate for Missing Middle and more generally a predictable, desired form, is to regulate minimum lot widths for each of the Missing Middle types. See examples of minimum lot sizes for each of the types in [chapter 5](#).

Assess the typical lot widths in pre-1940s neighborhoods and ensure that these types are allowed on one typical lot, in areas that have been identified as having Missing Middle Housing potential. Do not require a developer to aggregate two lots, which is typically not economically feasible.



Remove Open-Space Requirements

Open-space requirements are well intended but are very suburban in their intent, and such requirements often prohibit good urban-housing types.

They also are the primary reason one sees really bad balconies added to multifamily projects. This is typically done to meet the open-space requirements for each unit. The recommended approach is to completely remove open-space requirements. This is possible in walkable urban places because the high-quality public realm takes the place of these private open spaces.

Map the Zoning District That Allow These Types More Broadly

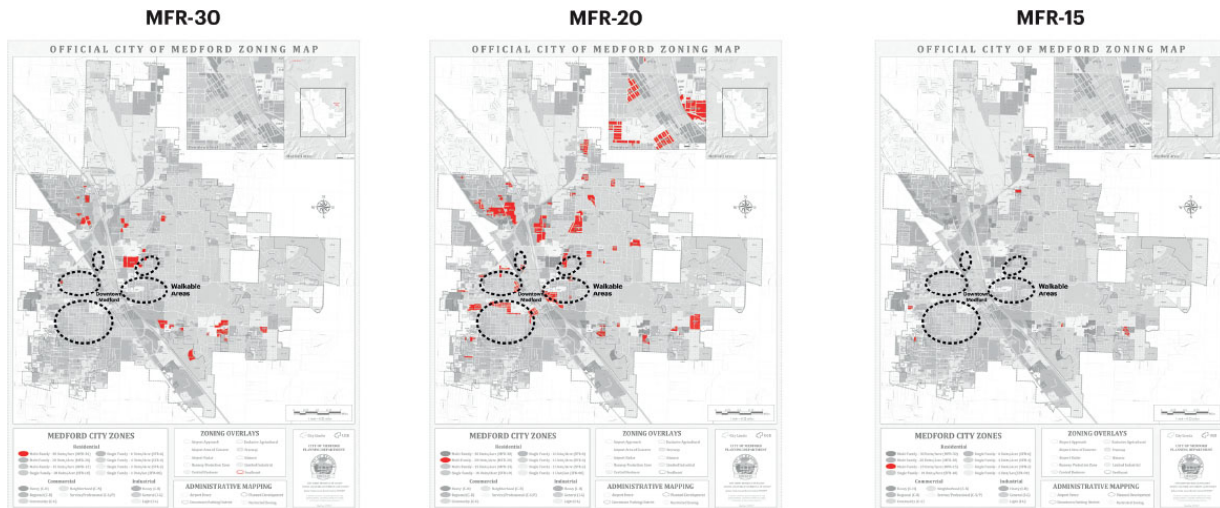
If effective zoned districts are in place or have been tweaked to enable Missing Middle Housing, these areas may only be mapped in a small geographic area or are not in ideal locations. The locations may be isolated, auto-dependent areas that are not very good areas for application of Missing Middle Housing. If this is a problem, identify walkable urban areas or intended walkable urban areas and prioritize them for application of Missing Middle Housing. This may mean converting additional single-family or commercially zoned areas to allow Missing Middle Housing. This may seem like a radical step for some jurisdictions, but, as further explained below, the City of Minneapolis and the State of Oregon have both taken dramatic steps to remove single-family zoning altogether, on a citywide and statewide basis, to enable Missing Middle Housing more broadly and directly.

Both lots are 6,000 sq. ft., but due to the different width along the street, different housing types or forms are appropriate for each. This is why minimum lot width is more important to regulate than lot size.

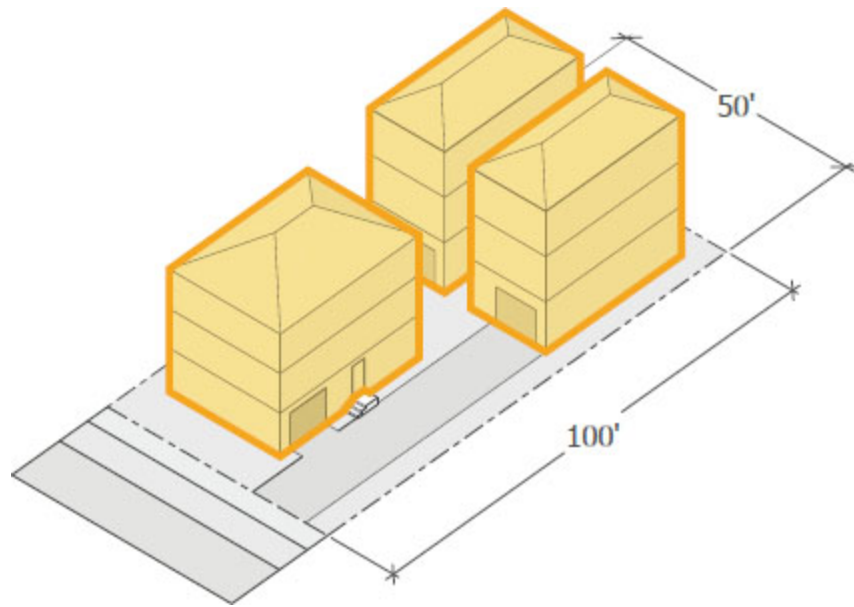
Do Not Allow Multiple Single-Family Homes on a Lot or Tuck-Under Town Houses

Be careful what is enabled. In existing medium-density zones, or when converting single-family zones to allow more than one unit per lot, unless it is specifically not allowed by the regulations builders will often build multiple single-family detached houses forced onto one lot or three-story, tuck-under town houses, without a strong relationship to the street. Why does this happen? Developers like these housing types because they can be sold as fee-simple units, meaning the buyer owns the land that the unit sits on rather than sharing ownership of the land as in a condominium with

stacked units; they eliminate or reduce construction defect liability, and there is an actual or sometimes perceived premium for these types. The problem is that these types are unlikely to deliver the range of affordability and attainability desired because they primarily deliver large units. They are also bad urban form, with one major flaw being that the three-story massing of the buildings is often pushed right up against the rear property line, impacting privacy of neighbors.



These analysis maps of Medford, Oregon, done as part of strategic zoning analysis and changes, reinforces the disconnect between where multifamily zones were mapped (in red) and where walkable contexts (black dashed lines), which are the ideal locations for Missing Middle Housing, exist.



In most cities, developers will build high-end tuck-under town houses or single family homes on lots zoned for multiunit buildings unless they are not allowed. To deliver a broader range of affordability and better urban form this should not be allowed.

CREATE NEW MISSING MIDDLE HOUSING ZONES

As a start, tweaking existing zones is often an easier fix, but does not achieve the level of predictability that creating new zones will, so the tweaks should be seen as a temporary fix or test, with the zone replacements as a longer-term goal. With this in mind, every zoning code should have a minimum of two zones to enable Missing Middle Housing: one to enable the lower and middle range of housing types from the Missing Middle spectrum, such as duplexes, cottage courts, and fourplexes (mostly two to two and a half stories maximum) and another to enable the upper end of the Missing Middle spectrum, which are slightly bigger buildings, such as courtyard apartments and multiplex: medium, with three to four stories and deeper buildings being allowed or encouraged. It is not always necessary but can be ideal to have a third zone at the upper end of what are now a city's single-family zones that enable the smaller-scale Missing Middle types, such as duplexes and cottage courts, along with single-family homes.

With housing affordability being a growing issue across the country, more and more cities are discussing the replacement of single-family zones with zones that allow a range of Missing Middle types that will be compatible in scale with single-family houses. For this approach, it is recommended that the “Parameters for Zone 1: Missing Middle Housing Zone” below be followed in creating that new zone, unless a higher degree of change or transformation has been determined in policies to completely change the form and scale of a neighborhood. In that instance only, the scale for “Zone 2: The Upper Missing Middle Housing Zone” defined below should be considered.



This zone-intent illustration from Cincinnati, Ohio's Form-Based Code illustrates the intended house-scale Missing Middle forms.

Whichever approach you take, here are some parameters for the zones you need to create and end up with after adjustments to your existing zones. Ideally a fair amount of documentation of existing conditions or local examples is completed to inform any adjustments that are needed to these numeric parameters.

Parameters for Zone 1: Missing Middle Housing Zone

- **Maximum building width:** 45–60 ft.
- **Maximum building depth:** 45–60 ft.
- **Maximum height:** two to two and a half stories
- **Max Density:** Ideally would not be used. If using density, 35 du/acre as a minimum threshold, ideally 65–70 du/acres maximum to truly enable a full range of attainable Missing Middle Housing units.
- **Allowed types:** Duplex: stacked, duplex: side-by-side, bungalow court, fourplex, courtyard apartment, multiplex: small, carriage

house, town houses (be careful about how many in a row you allow without a break)

- **Minimum lot sizes for each type:** See table for each type for general parameters that should be calibrated based on existing local conditions. Most of the Missing Middle types should be allowed on a 50 ft. wide lot, the bungalow court on a 100 ft. wide lot.
- **Ideally no off-street parking requirements:** Maximum 0.5 spaces per unit for one bedroom and larger, 0.25 spaces for a studio or unit under 650 sq. ft.
- Additional form regulations will be needed for bungalow courts to ensure good form.



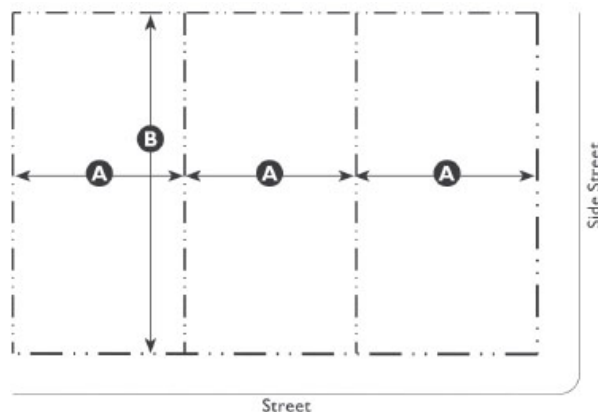
This zone-intent illustration from Cincinnati, Ohio's Form-Based Code illustrates the intended Upper Missing Middle Housing forms, which are taller and wider than the Core Missing Middle Housing forms.

TIP:

This zone should allow fourplexes on a 50 ft. wide lot and a cottage court with an unlimited number of one-story, 650 sq. ft. or smaller, cottages on a 100 ft. wide lot in a bungalow court format.

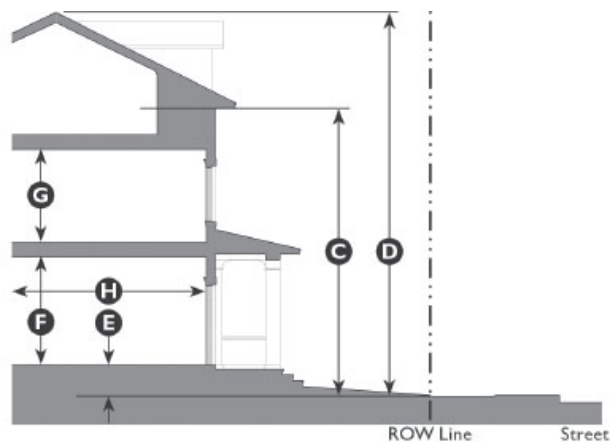
Parameters for Zone 2: Upper Missing Middle Housing Zone

- **Maximum width:** 65–85 ft. This is wider than Missing Middle zone, but not that much wider.
- **Maximum depth:** 5–10 ft. rear setback. Allows the building to go deeper into the lot than Missing Middle zones. This is one of the primary differences between Missing Middle and Upper Missing Middle.



Key

---- ROW / Lot Line



Key

---- ROW Line

C. Allowed Building Types

Building Type	Lot		Standards
	Width A	Depth B	
Carriage House	n/a	n/a	1703-3.40
Detached House:	30' min.;	75' min.	1703-3.60
Compact	50' max.		
Cottage Court	75' min.;	100' min.	1703-3.70
	100' max.		
Duplex	40' min.;	100' min.	1703-3.80
	75' max.		
Rowhouse	18' min.;	80' min.	1703-3.90
	35' max.		
Multi-Plex: Small	50' min.;	100' min.	1703-3.100
	100' max.		
Live/Work	18' min.;	80' min.	1703-3.130
	35' max.		

D. Building Form

Height		
Main Building		
Stories	2½ stories max.	
To Eave/Parapet	24' max.	C
Overall	35' max.	D
Accessory Structure(s)		
Accessory Dwellings	2 stories max.	
Other	1 story max.	
Ground Floor Finish Level	18" min.	
above Sidewalk		E
Ground Floor Ceiling		F
Service or Retail	12' min.	
Upper Floor(s) Ceiling	8' min.	G
Ground floor lobbies and common areas in multi-unit buildings may have a 0" to 6" ground floor finish level.		
Footprint		
Depth, Ground-Floor Space	24' min.	H
Accessory Structure(s)		
Width	24' max.	
Depth	32' max.	
Miscellaneous		

Loading docks, overhead doors, and other service entries shall be screened and not be located on primary street facades.

Some form-based codes integrate allowed Missing Middle Housing types directly into zoning standards to be more direct about the intended types and desired forms.

- **Height:** Three to four stories maximum
- **Allowed types:** courtyard apartments, multiplex: medium, multiplex: large
- **Minimum lot sizes for each type:** See table for each type for general parameters that should be calibrated based on existing local conditions.

USING A FORM-BASED VERSUS DENSITY-BASED APPROACH

If cities are taking the previous steps, with more thoughtful attention given to allowed scale/form/building envelope, your zoning/approach will automatically fall under a more form-based approach. A form-based approach is where the zoning more prescriptively defines a desired built form or allowed building envelope rather than primarily regulating with numeric parameters, such as density and FAR, within a larger allowed building envelope and allowing a broader range of built results within that allowed building envelope.

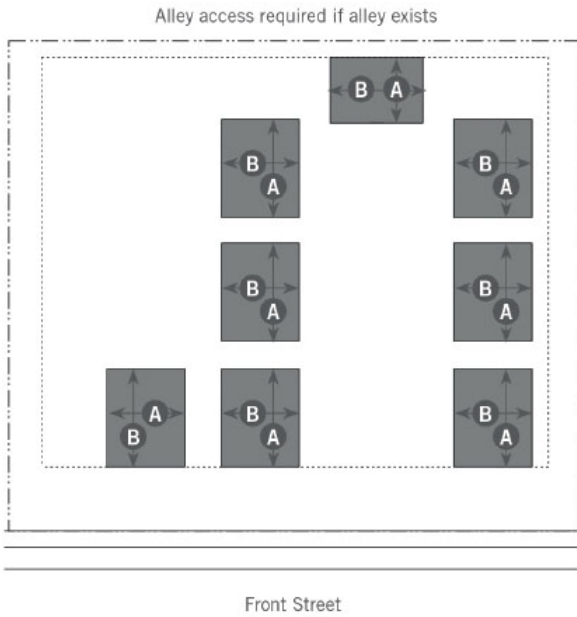
Most form-based codes do not regulate a density directly but still have a potential density output (which is calculated using a maximum number of units allowed within each housing type and minimum allowed lot sizes for each type to calculate the density, but the form intent, the building width, depth, height, and location on a lot, is regulated first and most importantly.)

Integrate Allowed Missing Middle Types for Each Zone

Not all form-based codes do this, but some take the regulations for Missing Middle Housing to the next level and directly enable an allowed range of building-type forms (not uses) within each new zone. This has been done in cities large and small, urban and rural, all across the country.

In this approach, each of the building types has a set of supplemental form standards that are often in a separate section of the zoning code, because each of the types apply to and are allowed in multiple zones. Clear referencing between these sections is important for usability. So the basic form parameters such as setbacks, heights, maximum building footprints,

and locations of parking are still established by the zone. Then supplemental form standards specific to a type reside in the “Building Type Standards” section. These standards sometimes trump the standards in the base zone standards. For example, a bungalow court might be allowed in a zone whose base standards allow two and a half stories. The bungalow-court standards typically trump that and set a height limit within a bungalow-court type at one and a half stories. Taking this height away is possible without discouraging development of this type because you are allowing developers to build more units on the lots. At first this may seem complex, but having these supplemental form standards by housing type is no different than having supplemental use standards like almost every zoning code does.



Key

---- ROW / Building Site Line ■ Building
 ---- Setback Line

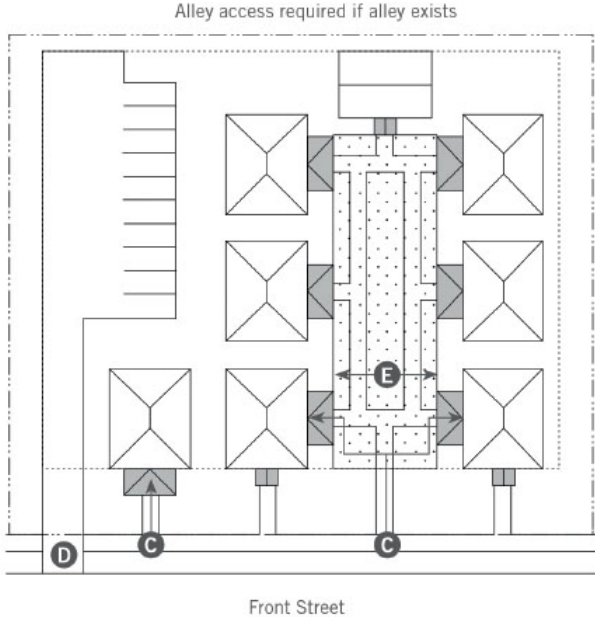
3. Building Size and Massing

Height	N-S	N-M
Max. Number of Stories	1	1
Max. Height to Highest Top Plate	14'	18'
Main Body		
Width	32' max.	A
Depth	24' max.	B

Facades on a street or civic space shall be designed in compliance with Section 40.14.080 (Specific to Massing, Facades and Architectural Elements).

4. Allowed Frontage Types

Porch: Projecting	40.14.090.D
Stoop	40.14.090.F
Dooryard	40.14.090.G



Key

---- ROW / Building site Line ■ Frontage
 ---- Setback Line □ Common Open Space

5. Pedestrian Access

Shared court shall be accessible from front street. **C**
 Main entrance location to units from shared court.
 Units on a corner may enter from the side street.
 Pedestrian connections shall link all buildings to the public ROW, shared court, and parking areas.

6. Vehicle Access and Parking

Driveway and parking location shall comply with standards in Subsection F (Parking) of the applicable zone. **D**

Spaces may be individually accessible by the units and/or a common parking area located at the rear or side of the building site.

7. Common Open Space

Width	20' min.	E
Depth	30' mins. (3 Units); 40' min. (5 to 9 Units)	

Required street setbacks and driveways shall not be included in the common open space area calculation.

An example of supplemental form standards needed to effectively regulate the cottage court type. Some cities do this through a Cottage Court Ordinance rather than a zoning district.

A word of warning: many recent codes that have attempted to take this approach of integrating housing types have failed because the numeric parameters are still not correct, or they still have a category called multifamily that is a catchall for all Missing Middle types and treat them more as a *use* rather than a *desired form*. Ultimately, it is the thoughtful metrics that are needed to enable Missing Middle, not just adding text about types into an otherwise conventional zoning code.

TIPS for Effectively Integrating Building Types into Your Zoning:

- If you take this approach, staff—especially staff at the front desk doing project review—will need some supplemental training about Missing Middle types. This approach is very different than they are used to because it takes the ability to understand physical form and housing types in a way and within a system they have not used.
 - If your code integrates allowed building types, one of them should not be multifamily. If it is, it is not truly enabling Missing Middle types partly because it is treating them all as one use (might as well call this medium-density residential) rather than a range of forms.
-

ENABLE MISSING MIDDLE BY REDUCING OFF-STREET PARKING REQUIREMENTS

The sad reality is that most cities of all sizes have done a better job housing cars than housing people. If the parking issue is not addressed by substantially reducing or removing off-street parking requirements, then Missing Middle will not be enabled effectively. Consider that, at a minimum, a wood-constructed garage adds approximately \$15,000 to \$20,000 to the cost of a unit, which can easily put the price out of reach for many buyers. If a city requires more than one off-street parking space for each unit, it is likely that it is not enabling Missing Middle Housing. In some higher-priced markets this requirement drops to 0.5 spaces per unit and often even zero spaces per unit.

The Best Parking Approaches

- Option 1: Ideally cities would remove all off-street parking spaces for residential uses, particularly in defined walkable urban contexts. In addition, cities would not require off-street parking for guests in either in these locations.
- Option 2: Remove parking requirements for residential buildings of less than eight units in walkable urban contexts.
- General note for best approaches: Allow on-street parking adjacent to a lot to count toward off-street parking requirements.

Backup Approaches (far less desirable)

- Regulate parking requirements by unit size or number of bedrooms so smaller units are not disincentivized. For example:
 - 0.5 spaces per studio or units under 650 sq. ft.
 - 1 space for one-bedroom units or units under 750 sq. ft.
 - 1.5 spaces per two bedrooms
 - 2.5–3 spaces per three bedroom and 1 space for every additional bedroom or
- Regulate by unit size as per above, and in this approach you could allow the inclusion of on-street parking adjacent to the site toward off-street parking requirements. This incentivizes Missing Middle on corner lots because of the extra curb space for corner lots.
- Pick a pilot-project area that requires no off-street parking for residential projects for up to fifteen units and then determine a set amount of time or number of built units after which it will be evaluated.

CASE STUDIES

HOW MISSING MIDDLE HOUSING IS BEING USED BY JURISDICTIONS, STATES, AND REGIONAL PLANNING AGENCIES

The following is a brief summary of some of the many and diverse types of Missing Middle applications that have been completed or are in process by cities, counties, and states across the country and around the world.

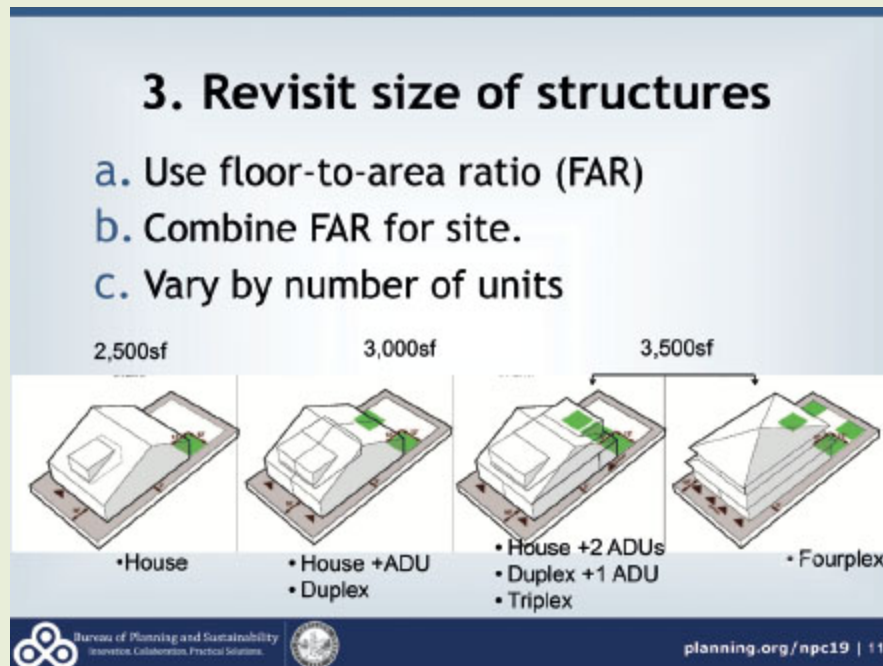
PORTLAND, OREGON

A Thoughtful, Effective Zoning-Reform Strategy

The City of Portland has one of the most well-thought-out and well-communicated (graphically, in text, and verbally) strategies in the country for removing zoning barriers for Missing Middle Housing, and it should be carefully studied. This effort was split into two tasks: task 1 was called the Residential Infill Project. This phase was intended to allow a broader range of Missing Middle types at the lower end of the Missing Middle spectrum, including small multiplexes, in areas previously zoned for single family; and, task 2 was called Better Housing by Design, which directly addresses the multi-dwelling zones (R3, R2, R1 and RH) outside the central city to allow a broader range of Missing Middle and Upper Missing Middle Housing types.

These efforts were started to address three concerns: 1. The rising cost of housing; 2. Increasing teardowns in neighborhoods; and, 3. The scale of additions and new buildings within neighborhoods.

Strategies for the Residential Infill Project were: 1. To limit the size of structures. This was to not allow further spread of oversized single-family homes; 2. Allow a broader range of housing types; 3. Thoughtfully increase the allowed FAR as the number of units increases. This was important to incentivize the construction of multiplexes on individual lots and disincentivize or not allow out-of-scale single-family rebuild or additions; and, 4. Improve the design of infill.



In Portland, Oregon, new regulations allow more FAR for multiunit buildings than for single-family homes to encourage Missing Middle Housing.

The City also conducted an economic feasibility analysis, which showed that with the introduction of new limits on structure size paired with allowing for more units in those structures, average rents were projected to decrease by more than half. In addition, as part of this comprehensive effort, the City completed a displacement impact risk analysis to inform the conversation about how these changes would impact different communities. This study demonstrated that the Residential Infill Project, if implemented, could substantially reduce displacement compared to a scenario where the Residential Infill Project was not implemented.

The goal of the Better Housing by Design is to establish development and design standards in Portland's multi-dwelling zones (R3, R2, R1 and RH) outside the central city to allow a broader range of Missing Middle and Upper Missing Middle Housing types and buildings that are larger than what would typically be classified as Missing Middle. The project will update the zoning code and align with the 2035 comprehensive plan to improve building design and development. This project is reviewing the zones on a citywide basis but also includes a special focus on East Portland to foster good new development that reflects the area's distinct characteristics and the needs of community members. These multi-dwelling zones are intended to accommodate up to twenty-three thousand new units by 2035 or approximately 20 percent of the growth.²

As part of the Better Housing by Design project, staff analyzed pre-World War II multifamily housing, such as courtyard apartment buildings, and found that most historic examples could not be built today in Portland's low-rise multi-dwelling zones because they exceed maximum density regulations. Proposed code amendments will regulate primarily by building scale, with more flexibility for numbers of units and new options intended to facilitate courtyard housing.

THE STATE OF OREGON

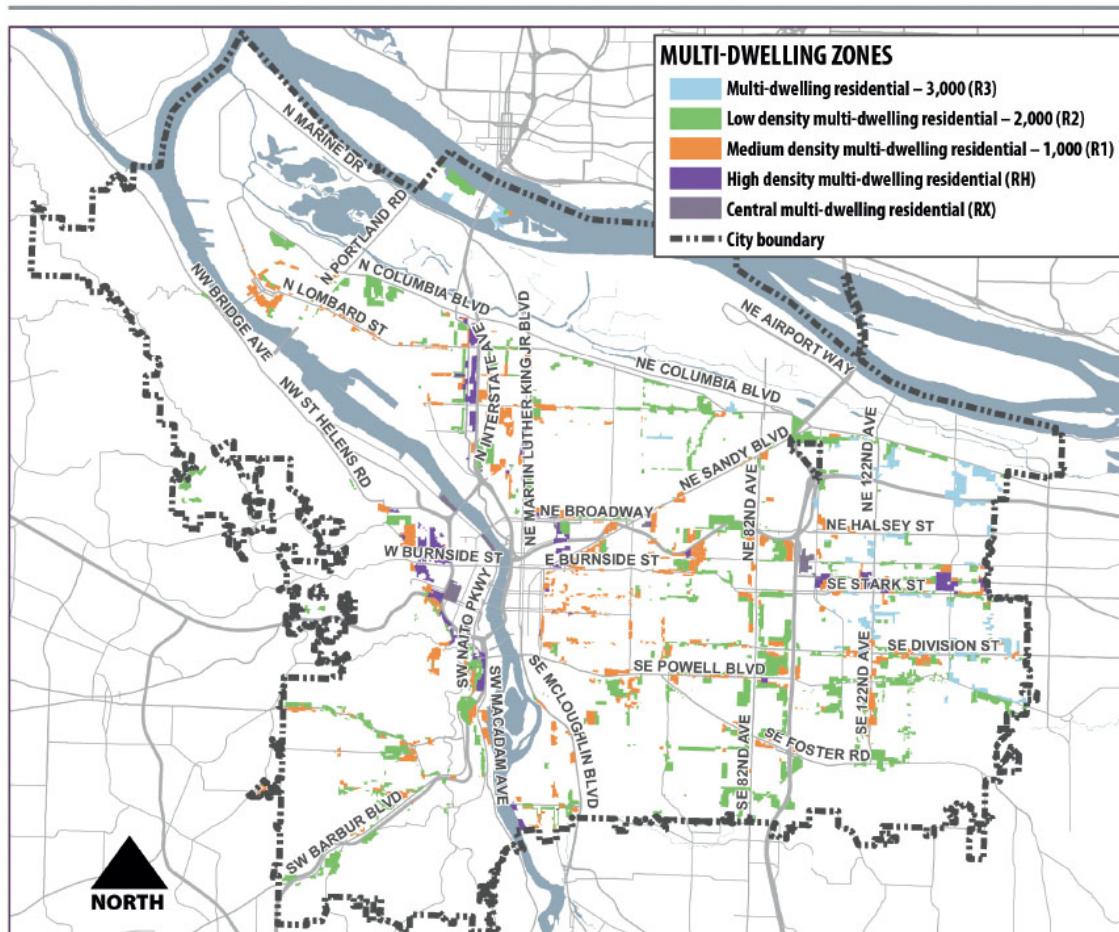
Statewide Enabling Legislation

In July 2019, Oregon's legislators passed the first state bill of its kind, Housing Bill 2001, which effectively eliminated single-family zoning statewide and enabled a range of Missing Middle Housing types everywhere to address its affordable-housing issues. The bill allows up to two housing units on all residential lots located within cities that have at least ten thousand residents. For cities of twenty-five thousand residents and up, the bill allows up to four units per lot, including row houses and cottage clusters. The bill will go into effect starting in 2020.

THE TWIN CITIES

Metro-wide Missing Middle Housing Policies

Over the past couple years, the Twin Cities metro area has become a leader in the discussion and application of progressive Missing Middle Housing policies, strategies, and zoning. Following is a summary of what has been done and is in progress.



Project schedule

Phase	Public Involvement – Comments/Testimony
1: Fall 2016 Research and Assessment	Existing Conditions Assessment Report available for public review end of December 2016 • Initial meetings and walks with community groups
2: Winter – Spring 2017 Concept Development	Public workshops on design concepts and preferred development outcomes. • Stakeholder working group meetings • Public open house on Concept Plan
3: Summer 2017 Code Development	Public review of Discussion Draft code amendments • Public open house • Stakeholder working group meetings
4: Fall 2017 – Winter 2018 Public Hearings / Adoption	Proposed Draft and Recommended Draft of code amendments available for public review • Public hearings at Planning and Sustainability Commission and at City Council



Better Housing by Design – February 2017

code-dev_better-housing 02/14/17

Map of Portland's multi-dwelling zones that illustrates how little of the city's geographic area allows multi-dwelling buildings.

MINNEAPOLIS

In 2009, the City of Minneapolis, Minnesota, took the lead in progressive Missing Middle Housing policies within its comprehensive plan. One of the overarching concepts reinforced by the plan's policy is "Affordable and Accessible Housing," which was reinforced by the goal that, "In 2040, all Minneapolis residents will be able to afford and access quality housing throughout the city." The plan states,

The loss of affordable housing units and changes in household income have resulted in a greater number of cost-burdened households—households in which more than 30 percent of household income goes toward mortgage or rental payments. Forty-nine percent of all households in Minneapolis are cost-burdened, but this is not equal across racial groups. Over 50 percent of black households and over 45 percent of American Indian and Hispanic households in Minneapolis are cost-burdened, whereas 1 in 3 white households are cost-burdened.

One of the steps taken to enable the City to achieve this goal within the plan was creating and approving "Policy 1—Access to Housing: Increase the Supply of Housing and Its Diversity of Location and Type." This policy allows up to three units (Missing Middle-scale housing) on any lot in the city, even those currently zoned for single-family housing in "neighborhood interiors farthest from downtown" and in "neighborhood interiors that contain a mix of housing types from single-family homes to apartments, increase housing choice and supply by allowing new housing within that existing range." This aggressive policy to remove single-family land use, and ultimately single-family zoning, from the entire city was groundbreaking and noticed across the country by policy makers, developers, planners, and community members.

After approving a comprehensive plan policy to allow up to three units on any lot previously zoned for single-family homes, the City of Minneapolis Community Planning and Economic Development department (CPED), in partnership with Minnesota Housing and Land Bank Twin Cities, initiated a pilot program to help fund Missing Middle-scale projects up to twenty units on city-owned lots. The City

committed \$500,000 to the first phase of this program. Total financing for ownership and rental projects is up to \$70,000 per affordable unit, and projects that demonstrate a compelling basis for deeper subsidy, such as including lower AMI restrictions, larger units, or longer-term affordability, may be eligible for up to \$95,000 per affordable unit. In addition, Land Bank Twin Cities offers interim construction financing to Minneapolis home developers subject to underwriting and approval. The City released an RFP for potential projects.

SAINT PAUL

Saint Paul, Minnesota, followed the lead of its sister city and in 2019 adopted a comprehensive plan that had Missing Middle Housing policy as a driver to address its current and future housing needs. Under “Goal 7: Strong neighborhoods that support lifelong housing needs,” one of the policies that directly supports Missing Middle Housing is Policy H-48. It expands permitted housing types in urban neighborhoods (as defined in the land-use chapter) to include duplexes, triplexes, townhomes, small-scale multifamily, and accessory dwelling units to allow for neighborhood-scale density increases, broadened housing choices, and intergenerational living.

CAPITOL AREA BASELINES

Baseline 1: The Capitol Area's Current Mix

Figure 7: Current Land Use in the Capitol Area

Minnesota's state capitol building is surrounded by neighborhoods filled with Missing Middle Housing types that are thoughtfully identified in the Capitol Area Plan document.

THE CAPITOL AREA PLAN

Even the Capitol Area Plan, led by the Capitol Area Architectural and Planning Board, which has authority over the geographic area surrounding the Minnesota state capitol, states, “Housing remains a key component of neighborhoods close to the Capitol, and a diversity in housing types, ownership, and residents will enrich the character of each residential neighborhood.” It then goes on to introduce Missing Middle Housing and states, “The existing low-rise range of housing types in Capitol and along Sherburne and Charles Avenues should be preserved and enhanced over time by the sensitive addition of new single-family homes and ‘missing middle’ types. As most of the structures in these areas are over 100 years old, a careful balance between redevelopment and restoration, renovation and rehabilitation will be required.” It then introduces policies and development standards to implement this objective and vision.

STATE OF MICHIGAN

Using Missing Middle as One Tool to Implement the State’s Place-Based Economic Development Strategy

The State of Michigan has advanced over the course of the past five to seven years from introducing the concept of Missing Middle Housing in its toolbox to inform strategic planning to targeting Missing Middle as project priority in its redevelopment incentive programs. “MIplace” is a place-based economic development strategy initiated in 2012 by the State. With an intensive academic literature/practitioner environmental scan, MIplace developed and operated a full-curriculum education component from 2013 to 2015, started the programming phase from 2015 to 2017, and since 2017 has integrated the principles within and as part of its community-development readiness certification.

According to Jim Tischler, now with the State’s Land Bank Authority, the primary goal of their work with Missing Middle Housing was not just to inform about the range of housing products but to create a paradigm shift within which local jurisdictions play an active role in the

development process. The State, very smartly, realized that this was important because it meant that local jurisdictions did not need to wait around for big developers to deliver results but rather could rely on incremental development and small builders to deliver much-needed housing. The State's development-readiness standards, marketed as "Redevelopment-Ready Communities," require locals to demonstrate inclusion and priority of Missing Middle in their policies, plans, and review processes to achieve certification and have priority status for State program investments.



This award-winning scheme in the State of Michigan's Missing Middle Housing Design Competition, submitted by Brian and Jennifer Settle, thoughtfully integrates courtyard housing behind a small mixed-use building.

In addition, the concept of Missing Middle has been adopted by the affordable-housing community statewide and is intrinsically linked to deliver workforce housing between 110 percent and 80 percent AMI and below, the latter being where federal subsidies come into play. Tischler stated, "The concept of Missing Middle Housing was a philosophical and a paradigm shift statewide that allows us to move toward production with socioeconomic integration of housing types at neighborhood scale. This would not have happened unless Dan had

created this concept and given his keynote presentation at the state APA conference. That planted the seeds of this effort.”

To promote the concept, in 2015 the State of Michigan also sponsored the first-ever Missing Middle design competition that looks at opportunity sites in various contexts around the state.

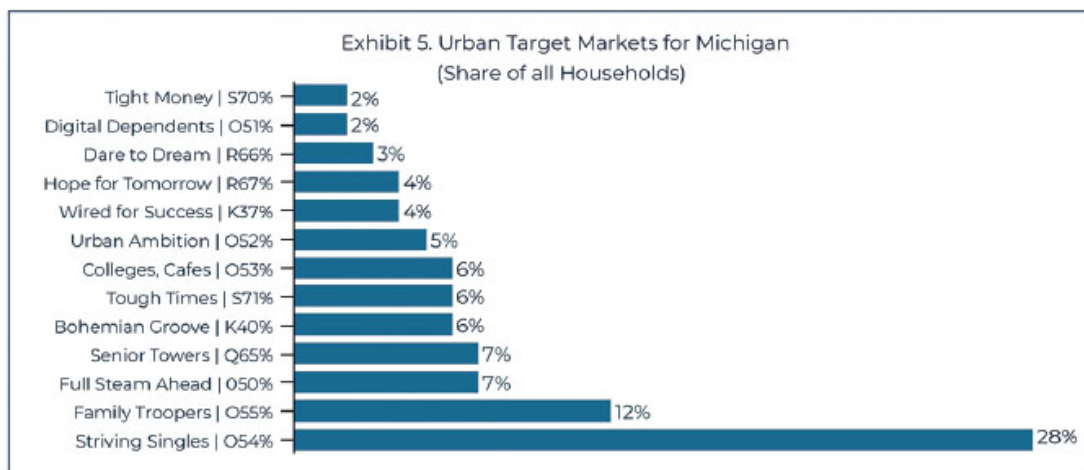
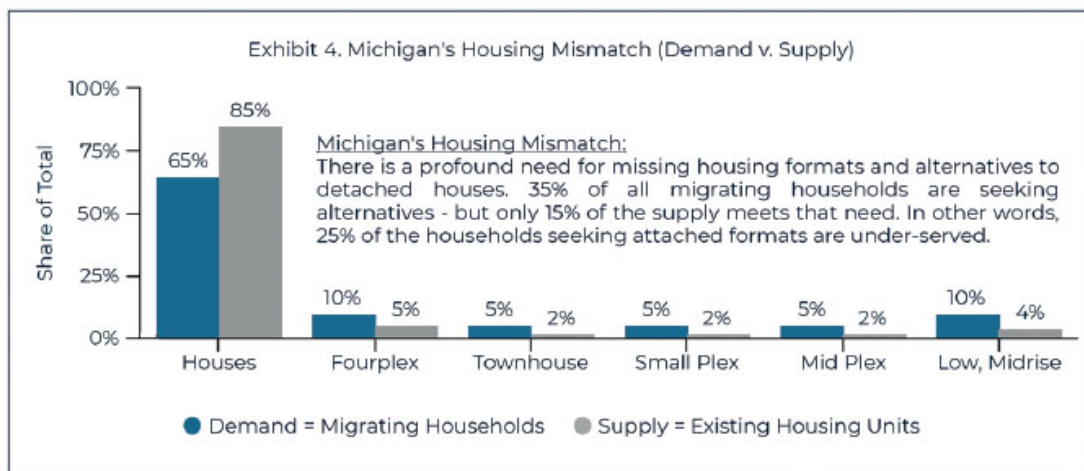
As part of the effort to enable Missing Middle, the State promoted the completion of a target-market analysis (TMA), a unique approach to residential-market analysis that specifically identifies the demand for each of the Missing Middle Housing types based on the lifestyle characteristics of migrating households. This is in contrast to a conventional residential-market analysis that identifies the demand for more general classifications of single family or multifamily and looks mostly in the rearview mirror at what types of housing have been built in the recent past.

This conventional approach to market analysis does not work to identify the demand for Missing Middle because none or little has been built in the recent past, thus making it easy to say it is not viable. According to Tischler, the primary goal for this effort was to cost-share with local jurisdictions to get this work done. This got them to commit and contribute to the effort, and then the TMA would provide them with data defining the need for these housing types in their specific jurisdictions. Tischler said, “this was especially important in the smaller, not-so-hot markets, where the government needs to assume a role in development to make it happen.” The State preapproved consultants for the target-market analysis and required one of these preapproved consultants to be used by local jurisdictions in order for this cost sharing to happen. The State cost-shared sixty-five TMAs, which seeded widespread adoption of the approach, and now over two hundred jurisdictions that have completed these TMAs.

What comes next? Tischler states, “Missing Middle Housing is so desperately needed in small towns, rural contexts, and larger cities, and the opportunities exist along corridors and in nodes as defined by the TMAs completed. We are pioneering use of parcel-specific ‘micro-TIFs’ for extraordinary-cost coverage, including purchase and rent subsidies, to accelerate production of Missing Middle units at incremental,

neighborhood scale. The delivery of Missing Middle Housing can drive the economic and social change that needs to happen across our state.” There is much to learn from this case study.

Michigan's Missing Housing - Here's the Scoop



Target-market analysis by the Incremental Development Alliance for the State of Michigan shows the gap between supply and demand for Missing Middle types and the oversupply of single-family homes.

ATLANTA, GEORGIA

The Missing Middle conversation has been percolating in the Atlanta region for several years now. In 2016, the Georgia Conservancy hosted a series of Missing Middle presentations that informed this conversation

and seems to have brought the need for these housing choices to the forefront of conversations about the future of the Atlanta region.

The City of Atlanta has made enabling Missing Middle Housing a priority within its citywide zoning-code update. Early in this process, the City's "Zoning Ordinance Diagnostic" document defined Missing Middle Housing as a priority, identified barriers and issues within existing zoning for the delivery of Missing Middle, and recommended strategies to remedy these barriers. Note that the City's communication was very effective, focusing on "ensuring housing diversity" and not focusing on density at all. The following is a list of the thoughtful recommendations included.

- **Recommendation 5.4: Building Types.** Define Missing Middle Housing building types during the update to the zoning ordinance. Types should include those historically found in Atlanta, such as duplexes and small apartment buildings, as well as those not traditionally found that serve current housing needs, such as town houses, cottage housing, and live-work units.
- **Recommendation 5.5: Integration into Existing Districts.** Allow Missing Middle Housing types within the appropriate existing or new zoning districts. Within existing districts this will require incorporation of the recommended building types and updated lot metrics. It will also require either increases to the permitted FAR or the complete elimination of FAR as a tool for controlling bulk.
- **Recommendation 5.6: R5 Amendment.** Amend R5 to require duplexes to resemble a single house. Typically, this will mean that the units must be stacked vertically or horizontally within a single building mass.
- **Recommendation 5.7: Land-Use Map.** Update the 15-Year Future Land Use Map to allow Missing Middle Housing in every medium-density residential (or equivalent) classification.

In January 2019, Atlanta's zoning changes were approved. These changes removed barriers for ADUs, made targeted changes to the metrics of the zoning to better enable Missing Middle, and created a

new zoning category called Multi-Unit Housing (MR-MU) that was specifically created to enable Missing Middle Housing.

TENNESSEE

NASHVILLE

Under the leadership of former director Rick Bernhardt, Nashville was an early adopter of the concept of a Missing Middle Housing. Their comprehensive plan, adopted in June 2015 has a housing element that identified Missing Middle Housing as a “key idea.” This policy continues to inform their small-area planning and form-based code applications that are implementing the plan. The plan states,

Missing middle housing can achieve multiple goals in Nashville’s housing market. It can be used to create transitions—in height, scale and density—between higher density centers and corridors and single family neighborhoods. When located by prominent corridors or commercial centers, missing middle housing allows more residents near transit, services, retail, and other amenities. Finally, missing middle housing can provide new housing options and meet the market demand for smaller units with less maintenance in walkable settings.... Changes to the zoning code to address transitions, contextual design, and ideal locations to permit the development of missing middle housing will aid in including this housing type in Nashville’s supply.

As part of its focused area-planning efforts, the City of Nashville continues to focus on enabling Missing Middle Housing. The City even named one of its efforts “The Missing Middle: Retrofitting the Centers and Corridors of Nashville.” This effort focused on defined focal points or centers as prime opportunities to enable more Missing Middle Housing to locate people closer to amenities such as transit and services. The effort will change zoning regulations, including height and bulk, to produce these house-scale Missing Middle types that will provide an

area of transition between centers or corridors and the existing neighborhoods.

CHATTANOOGA

In Chattanooga, the Chattanooga Neighborhood Enterprise, with support from the Lynd-hurst Foundation, led an effort to enable Missing Middle within the city's downtown-adjacent neighborhoods. These efforts included a design charette, a developer's boot camp, and the creation of a document that summarized a range of locally appropriate Missing Middle types, all created by the Incremental Development Alliance.

STATE OF CALIFORNIA

Local and State Efforts to Deliver the Much-Needed Housing in the State

California is one of the places that needs Missing Middle Housing the most, but local jurisdictions have lagged behind in their application of the concept. AARP's California chapter took the lead in providing information about Missing Middle Housing in California as part of its age friendly/livable community efforts by releasing a three-part series of Missing Middle Housing videos. Enabling the Missing Middle types was also an objective of many early form-based codes adopted in California cities, a few of which include Grass Valley, Ventura, Livermore, Petaluma, Hercules, Benicia, and Paso Robles.

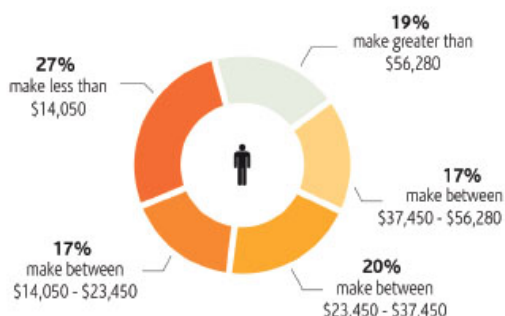


All Nashvillians need housing that is affordable ...

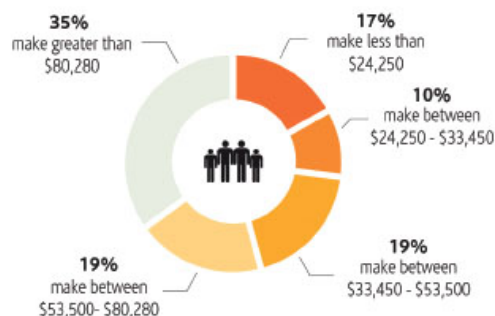
In recent years, people making less than 120 percent of Nashville Median Family Income (MFI) have felt the pinch of higher housing costs. This includes seniors and veterans on fixed incomes, those in the hospitality and retail industries, teachers and public safety providers, as well as nurses and technicians in the healthcare profession. Housing is considered affordable when a household spends less than 30 percent of their gross income on housing. Housing affordability leads to economic stability for individuals, families, and our community as a whole.

Extremely Low Income	Very Low Income	Low Income	Moderate Income
< 30 % MFI	30–50 % MFI	50–80 % MFI	80–120 % MFI

81% of individuals with income make less than \$56,280



65% of families with income make less than \$80,280



Sources:

Tennessee Department of Labor and Workforce Development, Labor Market Information, June 2014. Web Access April 2 2015.
<https://www.jobs4tn.gov/vosnet/Default.aspx>

Notes:

All data are derived from 2013 American Community Survey 1-Year Estimates for Davidson County TN and HUD FYI 2015 Income Limits. Family income limits in this case assume 1 Person (Individuals) and 4 Persons in a Family. HUD FY 2015 Nashville Median Family Income is reported at \$66,900.

In Nashville, households making less than 120 percent of median family income have been housing-cost burdened due to rising housing prices. The Nashville Comprehensive Plan uses Missing Middle Housing as one tool to deliver housing to these households.

There is a large list of housing-related bills recently passed or working their way through approvals. Some of these are quite progressive, such as the bill that started as SB827 and was brought back in a new iteration as SB50, both led by State Senator Scott Weiner, that would remove barriers for housing delivery by reducing local regulatory control, in particular to areas close to transit and jobs, by eliminating setbacks, density caps, and parking requirements and establishing much higher allowed height limits. Another, SB2, The Building Homes and Jobs Act, passed in 2017 as part of a fifteen-bill housing package aimed at addressing the state's housing shortage and high housing costs. The California Department of Housing and Community Development released \$123 million with approval of the Senate Bill 2 (SB 2, 2017) Planning Grants Program (PGP) to local jurisdictions to comply with the Objective Design Standards criteria defined in this bill. In addition, California's Department of Housing and Community Development (HCD) has substantially increased Regional Housing Needs Allocations (RHNA) for all jurisdictions, which will require local jurisdictions to think more creatively about delivering more housing, including Missing Middle scale. None of these efforts specifically address Missing Middle, but Missing Middle Housing will need to be one of the tools in every jurisdiction's housing toolbox to address these housing needs.

The following is a brief highlight of planning efforts that are directly utilizing Missing Middle Housing in the State of California.

THE GRAND BOULEVARD INITIATIVE

This effort facilitated the design of multimodal streetscape improvements for two case study locations on El Camino Real in Redwood City, California, and Palo Alto, California. The effort began in summer of 2017. As part of its "Policies and Programs" information this effort introduced and defined Missing Middle Housing and states, "Missing Middle Housing is a housing and urban design concept focused on moderate-density housing types that provide more housing choices while working within the scale and form of walkable neighborhoods."

THE CITY OF BERKELEY

The City of Berkeley's city council has approved funding to direct City staff to have a Missing Middle Housing study completed.

THE SACRAMENTO REGION

The Sacramento Area Council of Governments (SACOG) staff released a "Housing Policy Toolkit" that introduced the concept of Missing Middle Housing and included "Expand Missing Middle Zoning," as one of the recommended strategies for its jurisdictions. This is very relevant to this region as staff concluded through research that 90 percent of the land zoned for residential development in the region did not allow enough density to enable Missing Middle. As the State of California adjusts its Regional Housing Needs Allocations (RHNA), every jurisdiction is asked to accommodate more housing to address the state's housing crisis, Missing Middle is sure to play a role in enabling small, rural, suburban, and urban cities and counties to meet these required allocations/number of units.

COUNTY OF SAN DIEGO AFFORDABLE-HOUSING STRATEGY

The County of San Diego has been using the concept of Missing Middle Housing in its toolbox to deliver affordable-housing options.

In 2018 California passed AB 1637, also known as the Missing Middle Act. According to assembly member Todd Gloria, who introduced the bill, "AB 1637 authorizes the housing authorities in the City of San Diego and Santa Clara County to provide gap financing for middle-income housing projects—meaning, projects with at least 40 percent of units affordable to persons of low-income and 10 percent of units affordable to persons of middle-income. Currently, housing authorities can only assist with the low-income units of a project. With the Governor's signature, AB 1637 went into effect in 2018 and will operate as a pilot program with a four year sunset."³

This legislation focuses more on the affordability side rather than form and scale, but it is an interesting and important application.

Leadership Groups Promoting the Need in Regional Strategies

CASA was a group created by the San Francisco Bay Area Metropolitan Transportation Commission and Association of Bay Area Governments to help direct housing policy in the Bay Area to address the dire need for more affordable-housing solutions across the regions. This group comprised major employers, for-profit and nonprofit housing developers, labor and environmental leaders, public policy and affordable-housing advocates, transportation experts, charitable foundations, and elected officials. In late 2018 the twenty-one-member CASA Steering Committee finalized a ten-point set of policy recommendations document known as the CASA Compact. One of the seven core principles defined in this document was “Promote ‘Missing Middle’ housing types.” And one of the proposals was for minimum zoning near transit. According to the document, “This policy would create an inclusive mix of homes near transit and jobs, consistent with the goals of Plan Bay Area. It would spur the development of ‘missing middle’ housing types that are within reach of working families and blend into existing neighborhoods.”

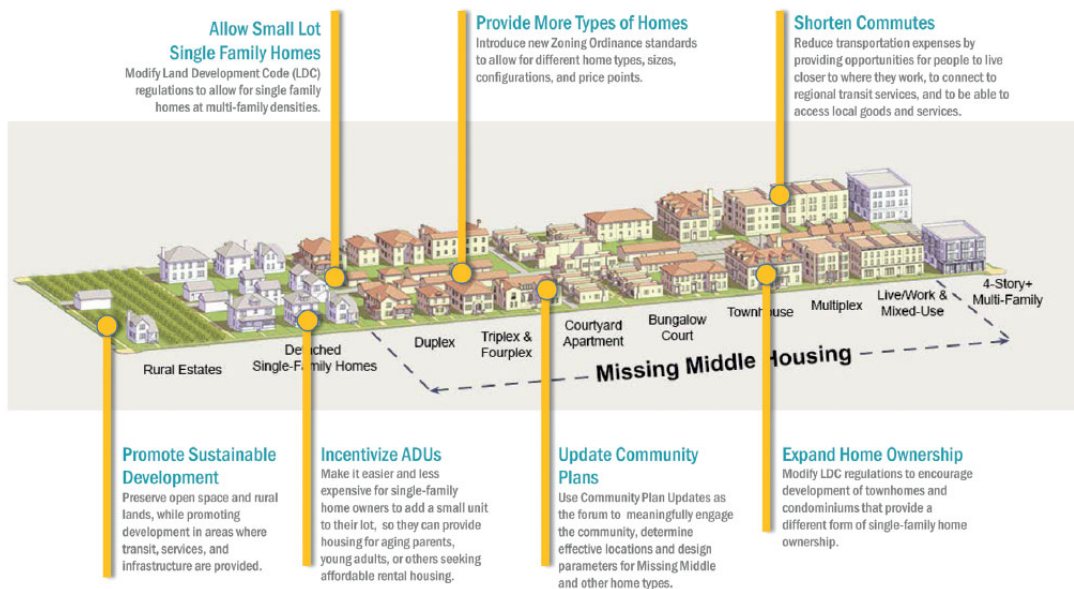
The one major mistake this document makes is that it refers to Missing Middle primarily as an income level rather than as house-scale buildings that can deliver affordability to middle-income households.

KAUAI COUNTY

Adapting and Applying Missing Middle for Rural Contexts

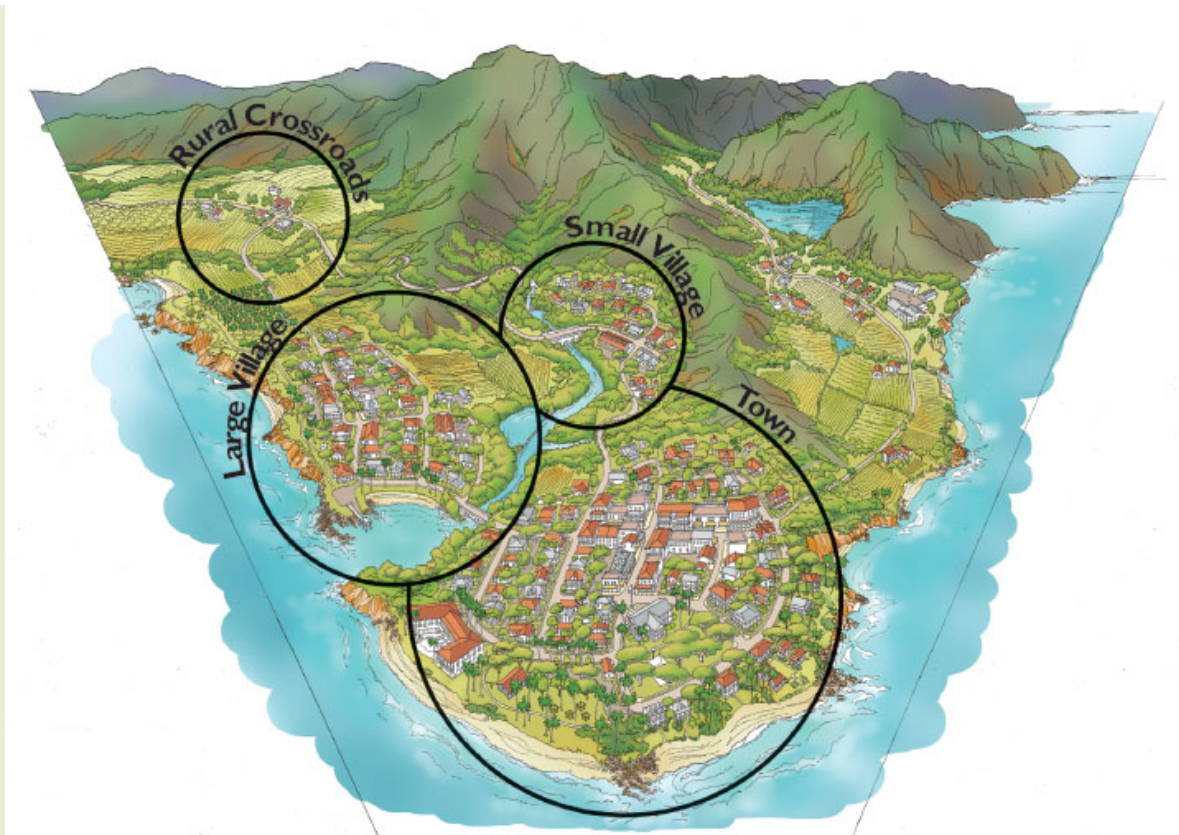
Hawaii’s Kauai County (the island of Kauai) has been working on several fronts to create more walkable, bikeable, and transit-friendly communities on the island and to expand housing choices, including Missing Middle Housing, under Mayor Bernard P. Carvalho’s Holo Holo 2020 sustainability initiative. This work included the adoption of a countywide multimodal land-transportation plan, a countywide comprehensive plan update, which won a National APA Burnham Award, and work on community plans for Lihue and South Kauai, that were implemented with a form-based code.

Strategies for Housing Affordability

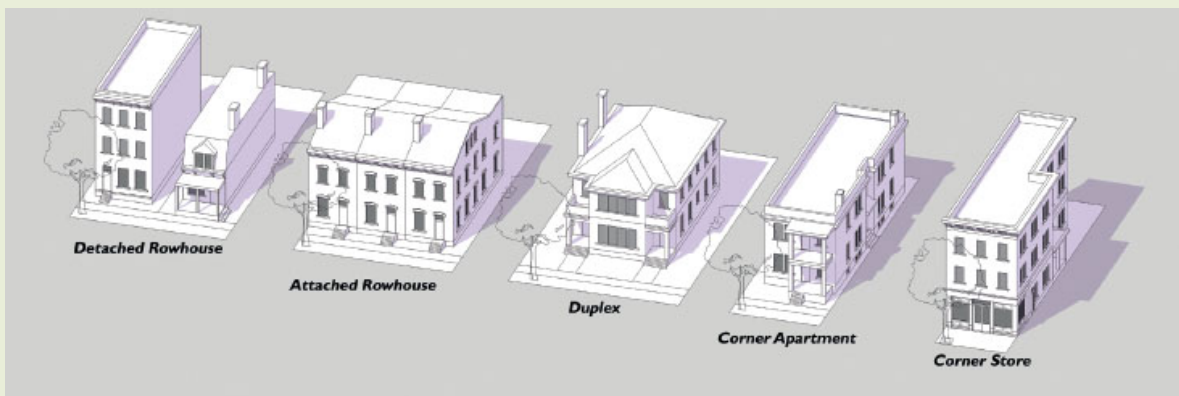


San Diego County in California is utilizing Missing Middle Housing as an affordable housing strategy.

A straw poll conducted by the County in June 2014 identified walkability and affordable housing as key concerns among county residents. The poll found that 37 percent of community members felt unsafe when walking in their community and 27 percent said housing is too expensive—only a quarter of the island’s residents can afford most of the housing inventory available. The South Kauai Form-Based Code was seen as a “pilot” to ensure development patterns can improve walkability and increase affordable-housing stock while preserving the county’s rural character.



The award-winning Kauai General Plan used a place-type approach that had Missing Middle Housing types as building blocks for these context-appropriate place types.



Cincinnati's strong history of Missing Middle Housing provided a solid foundation for their form-based code and comprehensive plan.

“Our area’s beauty and natural setting attracts worldwide interest, which is reflected in its high housing costs and resort-style development proposals,” said Marie Williams, long-range planner with the Kauai County Planning Department. “The South Kauai Form-Based Code will

encourage appropriately scaled infill development and ensure new greenfield development provides a variety of housing types. We hope this will translate to workforce housing adjacent to Kauai's largest resort center."

Within the comprehensive plan's "Actions by Sector," the plan defines infill housing as a priority. It then goes on to discuss supporting infill development and Missing Middle Housing and states.

Infill development is an important opportunity to diversify Kauai's housing stock through 'Missing Middle' housing in walkable town centers. 'Missing Middle' housing is characterized by small-scale, multi-unit housing types such as duplexes, fourplexes, bungalow courts, and mansion apartments that are not bigger than a large house, often integrated into blocks with single-family homes. Enabling this environment means creating or adapting planning and zoning requirements in a manner that will stimulate private investment in new or renovated structures.

The plan then goes on to establish a policy within the "Permitting Actions and Code Changes" section that states island housing officials will "substantially increase the amount of market rate multifamily and 'missing middle' housing on Kaua'i by requiring housing type diversity in all new subdivisions."

The larger-scale framework for enabling this infill and walkability was a place-types-based land-use framework in the island-wide comprehensive plan within which building types become the building blocks of the spectrum of walkable places defined.

CINCINNATI, OHIO

Building Upon a Long History of Missing Middle Housing Types to Promote Urban Revitalization

Cincinnati was also an early adopter of the concept of Missing Middle Housing, with a citywide form-based code that played an instrumental role in the framing of the concept of Missing Middle. Cincinnati has a

tremendous opportunity. In its urban neighborhoods they already have what other cities want and are trying to build: a variety of urban housing types, including some of the best Missing Middle Housing in the country; a network of neighborhood main streets ready to be revitalized; a rich, diverse, and well-built collection of historic architecture; and easily accessible open-space networks created by the topography weaving throughout these neighborhoods.

The Queen City positioned itself to capture this demand and put a strategy in place to make these neighborhoods complete places with everything urban neighborhoods have to offer. A comprehensive plan update, a citywide form-based code, and a completed streets document were created to enable the city to achieve these goals. A Sustainable Communities Challenge Grant from the Department of Housing and Urban Design funded the form-based code project. The Cincinnati Form-Based Code was adopted in 2013 and applied to four pilot neighborhoods, which were four of the forty-two defined walkable urban neighborhoods in the comprehensive plan that are all targeted locations for future application of the form-based code. These efforts were championed by Roxanne Qualls, the former mayor and vice mayor and city council member at the time.

AUSTRALIA

Spreading the Missing Middle Application Internationally

To deal with rapid growth and rapidly rising house prices, various jurisdictions in Australia have implemented the concept of Missing Middle Housing, including the Cities of Brisbane and Adelaide, and the States of New South Wales and Queensland to help them provide a more diverse range of housing choices at attainable price points.

NEW SOUTH WALES

“The Low Rise Medium Density Housing Code”

A report titled *A Place for Everyone*, created by Pricewaterhouse Coopers in 2017, points to estimates that greater Sydney will grow from 4.8

million people to about 8 million—the size of London—in the next forty years, creating a need for about 664,000 new homes across the next twenty years.

“So an increase in density isn’t optional—it’s unavoidable,” the report says. “But densification does not necessarily mean more high-rise apartments. If we get densification right, we have the opportunity to address both housing affordability and what’s referred to as the ‘missing middle,’ a diverse array of housing options to meet the needs of a diverse community.”

On July 6, 2018, “The Low Rise Medium Density Housing Code,” created by the planning department of the State of New South Wales in Southeast Australia, went into effect and is operating within eighty-two councils across the state. “The Low Rise Medium Density Housing Code” allows well-designed two-story versions of the following Missing Middle types: dual occupancies (duplex), manor houses (fourplexes and small multiplexes), and terraces (town houses) to be carried out under a fast track complying development approval. This approval can be as quick as twenty days if the submittal meets the State’s environmental planning policies.

REGIONAL PLAN FOR SOUTHEAST QUEENSLAND

Shaping SEQ

This regional plan for Southeast Queensland includes Brisbane and surrounding local governments, the Gold Coast, the Sunshine Coast to the north, and Toowoomba to the west. This plan does a very effective job of introducing the concept early in the document and then reinforcing the concept through policies, actions, and implementation steps throughout.

T4 Neighborhood Medium Footprint (T4N.MF)

1703-2.60 T4 Neighborhood Medium Footprint (T4N.MF)



A. Intent

To provide variety of housing choices, in medium footprint, medium-density building types, which reinforce the walkable nature of the neighborhood, support neighborhood-serving commercial adjacent to this Zone, and support public transportation alternatives. The following are generally appropriate form elements in this Zone:

Detached

Medium-to-Large Lot Width
Medium Footprint
Medium-to-Large Front Setback
Small-to-Medium Side Setbacks
Up to 2½ Stories
Elevated Ground Floor
Primarily with Stoops and Porches

B. Sub-Zone(s)

T4N.MF-Open Zone (T4N.MF-O)

The open sub-zone provides the same building form but allows for a more diverse mix of uses.

General note: The drawing above is intended to provide a brief overview of this Transect Zone and is illustrative only.

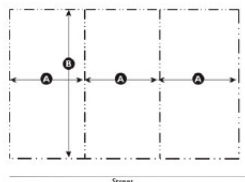
City of Cincinnati Form-Based Code

Final Draft 2/15/13

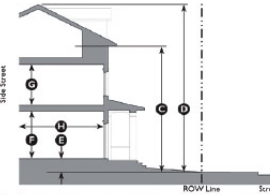
2-17 2-18

Final Draft 2/15/13

City of Cincinnati Form-Based Code



Key
--- ROW / Lot Line



Key
--- ROW Line

C. Allowed Building Types

Building Type	Lot		Standards
	Width A	Depth B	
Carriage House	n/a	n/a	1703-3.40
Detached House: Small	50' min.	80' min.	1703-3.50
Medium	75' max.		
Detached House: Compact	40' min.	80' min.	1703-3.60
Duplex	50' min.	100' min.	1703-3.80
	75' max.		
Rowhouse	18' min.	80' min.	1703-3.90
	35' max.		
Multi-Flex: Small	50' min.	100' min.	1703-3.100
	100' max.		
Multi-Flex: Large	75' min.	100' min.	1703-3.110
	100' max.		

D. Building form

Height	
Main Building	
Stories	2½ stories max.
To Eave/Parapet	24' max.
Overall	35' max.
Accessory Structure(s)	
Accessory Dwellings	2 stories max.
Other	1 story max.
Ground Floor Finish Level above Sidewalk	18" min.
Ground Floor Ceiling	8' min.
Service or Retail	12' min.
Upper Floor(s) Ceiling	8' min.
Ground floor lobbies and common areas in multi-unit buildings may have a 0" to 6" ground floor finish level.	
Footprint	
Depth, Ground-Floor Space	24' min.
Accessory Structure(s)	
Width	24' max.
Depth	32' max.
Miscellaneous	
Loading docks, overhead doors, and other service entries shall be screened and not be located on primary street facades.	

The Cincinnati Form-Based Code was calibrated to enable more of the broad range of Missing Middle types inherent in Cincinnati's historic neighborhoods. There were multiple now form-based zones. This zone allows a broad range of Missing Middle types, but requires larger front setbacks. This pattern was identified in the micro-scale analysis.

BRISBANE GROWTH STRATEGY

Plan Your Brisbane

The City of Brisbane, Australia, has a population of approximately 1.2 million and is growing at around 1,300 people a month. The city council wanted to undertake a significant community-engagement exercise to address this growth and particularly wanted to engage with sectors of the community that do not traditionally participate. The process consisted of a series of both traditional engagement exercises and then an online interactive game, called *Plan Your Brisbane*, which aimed to illustrate the impacts of different growth options for the player related to the key themes identified by the community. Players were given a "game board" with approximately six city blocks within which they had to

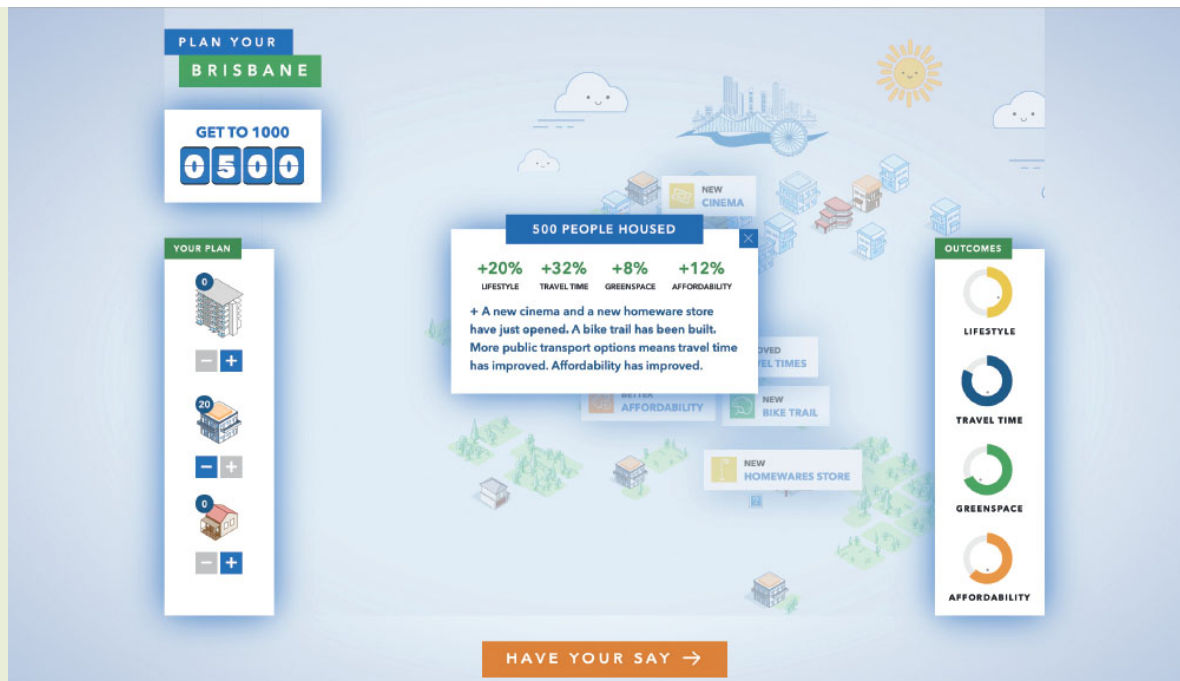
accommodate one thousand new housing units. The three scales of housing (on the lower left of the following image) that the player could choose from were large (mid-rise), medium (Missing Middle), and small (single family detached). The outcomes of lifestyle, travel time, green space, and affordability adjusted live based on the mix of housing types chosen by the player. For example, if more single-family housing was selected, travel times and green space increased and affordability decreased. In a scenario without single-family homes and a mix of Missing Middle and mid-rise housing, the outcomes for affordability are high, travel times are lower, and the lifestyle is medium, meaning that people may prefer to live in a different type but are willing to make the trade-off for the other outcomes.

The results were significant. City officials ended up with input from every suburb and every age group in the city, with one in five residents participating. All of that input was then aggregated and considered, and the council adopted a new corporate strategy known as Brisbane's Future Blueprint. As a next step, Brisbane is implementing the associated action plan, and one action is a housing strategy for the entire city, within which Missing Middle Housing plays a role in delivering more diverse housing choices and affordability.

ADELAIDE

The 30-Year Plan for Greater Adelaide

The 30-Year Plan for Greater Adelaide, which was first adopted in 2010 and updated in 2017, is a planning document intended to guide the development of the city. The recent update puts an emphasis on increasing residential densities through thoughtful infill, particularly with a focus on Missing Middle Housing types. The plan outlines a list of typologies that should be considered, including row houses, courtyard houses, granny flats, and laneway housing.

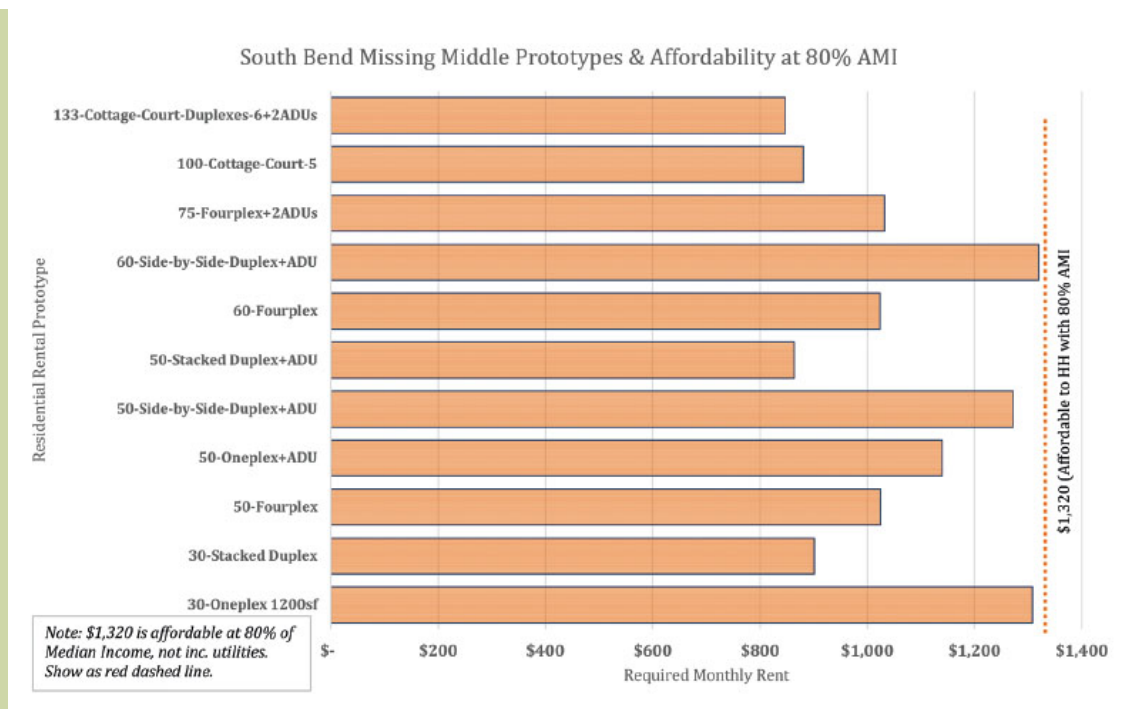


The interactive growth-scenario game used to engage community members in the discussion about trade-offs related to growth enables players to use single-family homes, Missing Middle, and high-rise buildings to test different growth scenarios. The outcomes adjust to show the impacts and benefits of the type of growth selected.

Box 7-3

Revitalization Strategies: Using Missing Middle Housing to Encourage Investment in Economically Challenged Neighborhoods

Missing Middle Housing was used as a strategy by the City of South Bend, Indiana, to help promote reinvestment in the Near Northwest Neighborhood, which had been in a steady state of decline for decades and had property values so low that it did not make economic sense to buy an existing single-family home and renovate it, because the cost of the renovation would ultimately be more than the value of the home. The local nonprofit the Near Northwest Neighborhood Association was the only group that was investing in the neighborhood, buying and repairing single-family homes and using federal subsidies to cover their cost gap and to sell them at affordable price points. The question this planning effort was intended to answer was: If the zoning was tweaked to enable Missing Middle Housing, would construction of these housing types make economic sense and promote reinvestment and infill in the neighborhood and still deliver affordable housing choices for existing and new residents? Or, would the City still need to provide economic incentives to cover the gap between cost and potential revenue through rents or sales of the units. Within the larger neighborhood, a twenty-eight-acre target site was selected by the City to complete a detailed Missing Middle build-out and to assess economic performance.



This proforma analysis by IDA illustrates that all of the housing types will generate housing affordable to households with 80 percent of AMI or higher with some types providing a higher level of affordability.

Existing conditions:

- A target-market analysis had defined a strong demand for the full range of Missing Middle Housing types in this neighborhood.
- 48 percent of the existing lots were vacant or the homes on them were abandoned.
- 82 percent of buildings were single-family homes. The remaining 18 percent were duplexes.
- Typical lot depth was 126 ft. with alleys.
- 32 percent of lots were less than 40 ft. wide, informing the need for small Missing Middle types that could fit on these narrow lots without lot aggregation.
- An additional 35 percent of lots were under 50 ft. wide.

The City of South Bend worked with Opticos Design to develop a palette of Missing Middle building types to study. The types ultimately included in the economic analysis were duplex: stacked, duplex: side-by-side, fourplex, cottage court, and townhomes. The ultimate build-out for the target area included 44 percent duplexes, 24 percent fourplexes, 18 percent cottage courts, and the rest a mix of Missing Middle types. A pro forma analysis was performed on each of these types by the Incremental Development Alliance, including options with accessory dwelling units (ADUs) in addition to the types on each lot. The pro forma analysis assumed no land cost, because the City of South Bend owned many of the lots, and a 7.5 percent return. With a ten-year build-out, there would be an estimated \$1.3 million annual tax revenue for the city. In addition, the pro forma analysis shows that newly constructed examples of

the full range of Missing Middle types will deliver affordable rents for households that are at 80 percent of the city's medium income and above, and the stacked duplexes and cottage courts performed the best in terms of being able to bring them online at the lowest rents.



Existing Condition



Phase I Build Out: 0-5 years



Phase II Build Out: 5–10 years



Phase III Build Out: 10–20 years

These drawings illustrate the incremental, locally driven stabilization and build-out strategy for the neighborhood that has been in a state of decline for over thirty years. Enabling Missing Middle Housing is the key to allowing private investment in the neighborhood.